

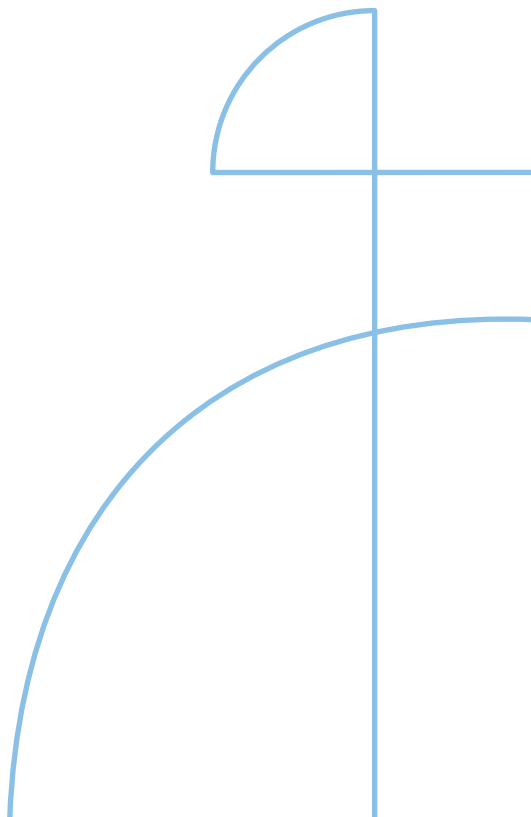
Doctoral Thesis in human-computer interaction

Ethical Reasoning in Tech Work

From Individual Responsibility to Collective Action

KRISTINA POPOVA

KTH ROYAL INSTITUTE OF TECHNOLOGY



Ethical Reasoning in Tech Work

From Individual Responsibility to Collective Action

KRISTINA POPOVA

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Abstract

Technology production is a complex process that requires the accumulation of resources and in which very little is done individually. Given the fragmentation of tech development and the limits of individual action, how much can tech workers affect the tech they produce? Do they care about the tech they are building and its implications for the world? Who should act to ensure that technology is "ethical"? What are the possibilities for individual and collective action that exist in tech workplaces? These are the questions I aim to answer in my dissertation.

This dissertation is an interdisciplinary project connecting four studies with the methods from design research, ethnomethodology and sociology. The papers are published within the fields of human-computer interaction (HCI) and computer supported cooperative work (CSCW) and positioned within the tradition of studying ethics in practice, which approaches ethics by studying the work practices in specific tech organisations. Theoretically, my work is grounded in the philosophical tradition of ethics of care with its focus on concrete situated actions instead of omnirelevant rules, in the ethnomethodological tradition of respecifying theoretical phenomena as matters of practice, and in moral anthropology with its focus on ethics as shaped by communities.

The four papers combined in the thesis explore the ethical reasoning of tech practitioners in academic settings, government officials working with AI in Sweden, and technology practitioners working in commercial companies in Europe. I rely on a variety of qualitative data: (1) a video-ethnography of design ideation sessions at the university; (2) an interview-study of ethical responsibility by tech practitioners in academia, governmental sector and the industry; (3) a a workshop-based study of voicing discomfort (critique) in a design workshop with academic; (4) an interview-based study of ethics discourses in tech com-

panies in Europe. The studies contribute to design and HCI with an empirical research of ethical reasoning in everyday design work.

The thesis aims to advance an anthropological, human-centred take on ethics in HCI. I argue in favour of including affect and emotion of tech practitioners into the discussion on ethics in HCI and emphasise the limitations of an individualistic take on ethics. Affect is important for connecting the level of everyday practices with structural power relations. Understanding the limitations of individual action and focusing on the potential for collective action instead is important for balancing responsibility with power. This thesis also draws attention to the inequality within the tech sector as a factor that hinders the possibility for forming a shared agenda and engaging in collective action.

Keywords
ethics, tech work, responsibility, emotions

Sammanfattning

Teknikproduktion är en komplex process som kräver ackumulering av resurser och där väldigt lite görs individuellt. Med tanke på fragmenteringen av utvecklingen av teknik och begränsningarna för individuellt agerande, hur mycket kan arbetare påverka den teknik de producerar? Bryr de sig om den teknik de producerar och dess konsekvenser för världen? Vem bör agera för att säkerställa att tekniken är "etisk"? Vilka möjligheter för individuellt och kollektivt agerande finns det på teknikarbetsplatser? Det är dessa frågor jag strävar efter att besvara i min avhandling.

Denna avhandling är ett tvärvetenskapligt projekt som kopplar samman fyra studier innehållande metoder från designforskning, etnometodologi och sociologi. Artiklarna är publicerade inom områdena människa-datorinteraktion (MDI) och datorstödd kooperativt arbete (CSCW) och positionerade inom praktisk etik, en forskningsansats som studerar etik genom att granska arbetsmetoder i specifika teknikorganisationer. Mitt arbete är teoretiskt grundat i den filosofiska traditionen i omsorgsetik med fokus på konkreta situerade handlingar istället för allmängiltiga regler, i den etnometodologiska traditionen att specificera teoretiska fenomen som praktiska frågor, och i moralantropologi med fokus på etik som formas av samhällen.

De fyra artiklarna som kombineras i avhandlingen utforskar det etiska resonemanget hos teknikutövare i akademiska miljöer, myndighetspersoner som arbetar med AI i Sverige och teknikutövare som arbetar i kommersiella företag i Europa. Jag förlitar mig på en mängd olika kvalitativa data: (1) en videoetnografi av designsessioner vid universitetet; (2) en intervjustudie om tillskrivning av etiskt ansvar av teknikutövare inom akademin, den offentliga sektorn och industrin; (3) en workshopbaserad studie av uttryck för obehag (kritik) i en designworkshop med akademiker; (4) en intervjubaserad studie av etiska diskurser

i teknikföretag i Europa. Studierna bidrar till design- och MDI-fälten med empirisk forskning om etiskt resonemang i vardagligt designarbete.

Avhandlingen syftar till att främja en antropologisk, människocentrerad syn på etik inom MDI. Jag argumenterar för att inkludera affekt och känsla i diskussionen om etik inom MDI och betonar begränsningarna med en individualistisk syn på etik. Affekt är viktigt för att koppla samman vardagliga observerbara arbetsmetoder med strukturella maktrelationer inom teknikssektorn. Att förstå begränsningarna med individuellt handlande och istället fokusera på potentialen för ett kollektivt handlande är viktigt för att balansera ansvar med makt. Denna avhandling uppmärksammar också ojämlikheten inom teknikssektorn som en faktor som hindrar möjligheten att forma en gemensam agenda och engagera sig i kollektivt handlande.

Nyckelord
etik, tekniskt arbete, ansvar, känslor

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I have learned during my PhD that research is never an individual enterprise, even if it appears so at times. Most of the contributors are not listed as authors, but I do think about all research as collegial. This one is not an exception.

Kristina Popova
Stockholm, April 7, 2026

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Introduction

1 Introduction

Technology development is a complex process. The people involved in the production of technology are geographically dispersed, culturally diverse, and located on different levels of organisational hierarchies. The production of an iPhone, for instance, requires coordination and collaboration of people across the globe, relying on global infrastructures and supply chains (Crawford 2021). Given the scale, the problem of distributing ethical responsibility for the outcomes of technology development is far from straightforward. Who is supposed to be responsible for the consequences of technology? Is it the user, the developer, the CEO, politicians, citizens, all of them? But if all of them are responsible, how exactly do they exercise this responsibility and put it into action? And what would be the limits of responsibility of an individual designer or an engineer working, for instance, on iPhone development or designing drone technology? These are some of the questions that inspired my studies on ethics in tech production.

While the ethics of technology is acknowledged to be important in the era of generative AI, self-driving cars, and drone wars, it remains abstract and remote from tech practice. This is understandable, given that ethics is originally a sub-discipline of philosophy — a discipline that deals with abstract theoretical constructs. Now, however, ethics is also a part of Human-Computer Interaction (HCI), Computer-Supported Cooperative Work (CSCW) (Nunes Vilaza et al. 2022), and design as fields particularly interested in developing more hands-on approaches to ethics and understanding how to make technology more ethical, less harmful, and more aligned with human values (Friedman and Hendry 2019).

In the field of design, various frameworks have been developed with the purpose of incorporating ethics into the design process. One of the examples is

Value-sensitive design (Friedman 1996) as one of the most well-known methodologies that aims to incorporate ethics into the design process. There are also traditions of critical (J. Bardzell and S. Bardzell 2013) and reflective design (Sengers et al. 2005) arguing for development of certain values in design as discipline. AI scholars are working on guidelines to support more fair and less harmful AI (Jobin, Ienca, and Vayena 2019).

One topic that I see as requiring more attention in the conversation about ethics across disciplines concerns the translation of these recommendations into action outside the academic domain. Who is supposed to make sure that technology is 'ethical' and determine what being ethical means? And what would be the role of people involved in the production of technology – tech workers? I find these questions important both practically – to transition from discussion about better tech to action – and theoretically, since they are asking to define the responsibility of tech creators.

Within HCI and CSCW these questions are asked within the studies of ethics in practice (Wong, Madaio, and Merrill 2023; Greene, Hoffmann, and Stark 2019; Widder, Zhen, et al. 2023; Shilton 2013; Boyd and Shilton 2021; Colin M. Gray and Chivukula 2019). This tradition sets out to explore how industry professionals, engineers, and designers of technology approach ethics in their work: instead of studying normative ethical principles, the scholars of ethics in practice, look at the norms and guidelines as they in the context of specific organisations and professions (Wong, Madaio, and Merrill 2023; Madaio et al. 2020). The researchers analyse, how specific values get translated into technical solutions (Shilton 2013; Colin M. Gray and Chivukula 2019), look at how tech practitioners deal with ethical tensions in their work (Figueras, Verhagen, and Cerratto Pargman 2022) and collisions between ethics and business goals (Lindberg, Karlström, and Männikkö Barbutiu 2021).

A similar practice-oriented approach to ethics in Design research is the study of in-action-ethics (Frauenberger, Rauhala, and Fitzpatrick 2016) and micro-ethics (Spiel et al. 2018), examining ethics in the details of Design researchers' work rather than on the level of institutions or theoretical principles. They offer a conceptual shift from understanding ethics as matters of universal judgment to context-dependent decision-making in design practice within academic settings.

My thesis contributes to practice-based research on ethics within these empirically and pragmatically oriented traditions by looking at everyday practices of tech practitioners in three kinds of tech workplaces. The empirical settings I am looking at vary from my own university environment, where HCI and Design researchers build prototypes and create concepts that may affect development of tech in the future, to software engineers and UX writers working in government agencies in Sweden and tech industry in European countries.

The collection of papers combined in this thesis approaches ethical reasoning

as an observable phenomenon that can be studied empirically in the everyday work of tech workers. By doing so, I am inspired by the ethnomethodological tradition of respecifying theoretical phenomena as phenomena of practice (Francis and Hester 2016), ethics of care with its emphasis on the contextual nature of ethical reasoning (Tronto 1993), and moral anthropology with its interest in people as ethical actors (Zigon 2007). My interest lies in understanding how technology practitioners themselves strive to ensure that their actions are ethical, what they choose to care about and pay attention to, and how they explain to themselves and others why what they do is ethical.

I suggest that looking at everyday practices and discourses about these practices can be useful to see how the everyday incorporates or ignores theoretical problems from the ethics of tech, for instance, the question of building technology not harmful for users. One reason for that is to recognise the importance of everyday practices in contrast to the realm of institutional politics. How do we study ethics in practice as an observable phenomenon and a part of everyday work of ordinary tech practitioners – those who are not CEO, heads of research institutions and governmental agencies? This thesis is a collection of papers that take different stances to approach ethics in practice and study ethics as a part of tech work.

1.1 Research Questions

This thesis studies how tech workers approach ethics in their work. A key reason for asking this question is my interest towards tech workers reasoning around 'the problem of many hands' (Figueras 2025), i.e. the difficulty with reconciling the scale of technology creation and the limitation of individual responsibility. I am interested in the problems caused by these discrepancies and how tech practitioners solve them.

I study ethical reasoning in tech work by focusing on workplace communication and the workers' emotions. I argue that paying attention to emotions, which in my studies I approach discursively as people talking about their feelings and using affect in their descriptions, is important because it helps us to understand the limits of individual action and the (limited) potential for collective action and distribution of power in technology workplaces.

In this kappa I connect the level of everyday workplace interactions with broader theoretical questions in the ethics of tech, e.g. how do we build technology that is not harmful and technology that improves human lives[1]. My studies aimed to understand whether these big questions, important to tech ethicists, surface in technology workplaces at all. And, if they do, in what way, under which circumstances and with which results? All these sub-questions are compressed in

[1]What is meant by improved and how we see harms do not have universal answers. I see these questions as matters of values. From my standpoint, acting ethically in tech means promoting equality and eliminating oppression.

my first research question, formulated in the language of ethics of care (Tronto 1993):

1.1.1 RQ1: What do tech workers care about in their work and their workplaces?

By asking this question in the context of studying ethics, I rely on the tradition of ethics of care as a philosophical theory that approaches ethics in context of practical activities and focuses on acts of responsibility expressed in actions rather than abstract rules and rights (Tronto 1993, pp. 79).

Built into this question, there is a distinction between *caring about*, which means recognizing the needs of a specific group, and *taking care of*, which means responding to these needs through actions. Asking what tech workers care about and what they take care of means asking whose needs do they recognise and whose needs do they act upon in their workplaces. Consequently, it also means asking, whose needs are not recognised and not acted upon.

1.1.2 RQ2: To whom do tech workers attribute ethical responsibility in their work?

The second question is based on the premise that tech workers are already familiar with some of the concepts from the discussion on ethics of technology, for example, they may have an idea that technology is not supposed to be harmful or biased. They may also have an idea of the extent of their personal responsibility for ensuring that the tech they build is not harmful, and an assumption about who else is responsible for exercising this ethical responsibility.

1.1.3 RQ3: What are the possibilities and limitations for collective actors in tech workplaces?

One of the critical assumptions I had even before doing the studies was that individual responsibility sits very problematically with the complexity of tech development. If an individual agency in tech is limited, as well as the realms of individual responsibility, then how is it possible to transition from seeing ethical problems as matters of individual decision making to matters of collective action? Are there possibilities for ethics of tech to become a part of the collective agenda of tech workers, who could engage in a collective stride towards creating a more ethical technology? With the question 3, I want to look into the possibility of regulating technology from inside the tech sector, not from the top-down through regulations and corporate policies, but from bottom-up – as when Google had to shut down its drone project Alphabet in 2017 because of the pressure from its employees (Burgees 2017).

1.2 Theoretical Approach

I approach ethics as the study of observable phenomena rather than as theoretical arguments alone. In this, I have several theoretical inspirations. They all overlap in terms of turning their gaze towards situated practical actions rather than universal theoretical considerations.

The first inspiration comes from ethnomethodology and its tradition of 'respecifying' theoretical (sociological) phenomena as matters of practical actions (Francis and Hester 2016; Garfinkel 2002). Some of the examples here are studying scientific knowledge through practices of scientists working in the lab (Lynch 1993; Livingston 2008; Sormani 2014), gender as performing gender or mathematical reasoning as a situated bodily activity (Livingston 2008). In Chapter 3 I I will present the theoretical components that are needed in order to better understand the premises behind my studies, describing some principles of ethnomethodology to explain how respecifying ethics can happen and why it is important.

My second theoretical ground is the philosophical tradition of Ethics of Care (Tronto 1993; Gilligan 1982; Noddings 1984; Held 2006). The main argument of ethics of care theorists was to start looking at practical circumstances, where ethical decisions are made, rather than looking only for abstract ethical rules applicable in any situation, which would be, for instance, Kantian approach based on universal reasoning or the principle for ethics of justice (Rawls 1958; Rawls 2005)[2]. They have also suggested looking at the work of care and the domain of private life as the domain where morality is practiced, instead of just looking at public life as the sphere of production (Tronto 1993).

There are 3 main principles in the ethics of care, which are also important for my work:

- focus on responsibility instead of rights and rules;
- focus on concrete circumstances rather than abstract principles;
- focus on actions, 'the acts of care' and not on principles (Tronto 1993).

Care ethics theorists call to formulating ethical principles as an inference from the acts of caretakers. Care is understood as an "activity to maintain, continue, and repair our world so we could live there as well as possible" (Tronto 1993, p. 103, citing Bernice Fisher). The empirical and contextual orientations of the care ethics, as well as its focus on work, i.e. practices rather than intentions, and responsibilities rather than rights are the key theoretical underpinnings of my work. The relation of ethics of care to other ethical theories, as well as the role that it plays in my work, is mostly discussed in Chapter 1.

[2] Ethics of care is not against ethics of justice, but it is not limited to the concept of rights only.

The third inspiration comes from the moral anthropology of Zigon (Zigon 2007), which is especially important for the empirical part of my work. Moral anthropology is concerned with ethical reasoning in specific moments and exercised by people. Zigon studied moral decision-making in situations of uncertainty, the moments of breakdown, when previously taken-for-granted rules were no longer applicable (Zigon 2007). His study took place at the Post-Soviet space after the collapse of the Soviet Union, when the communist ideology was replaced with free market (Zigon 2007). I do not look at tech ethics from a historical perspective, but I look at the moments of 'breakdown' and uncertainty as emotionally charged moments when moral norms clash with each other and routine behaviours are disrupted.

Within the domain of HCI and CSCW, these traditions are close to the tradition of studying ethics in practice, which contrasts the study of specific tech organisations and tech workers to the practice, more familiar for tech disciplines, of formulating guidelines and principles (Figueras 2025; Figueras, Verhagen, and Cerratto Pargman 2022; Greene, Hoffmann, and Stark 2019; Madaio et al. 2020; Widder and Nafus 2023; Powell et al. 2022; Wong 2025).

1.3 The studies

The thesis is based on four empirical studies, where I study ethics in the work of different groups of technology workers: academic HCI researchers; industry tech workers – software engineers, UX designers, product managers and similar professionals employed in different sectors in Europe; and AI experts working for governmental agencies in Sweden.

These four studies come from two different research projects and employ diverse methodology as a combination of qualitative semi-structured interviews (Paper 2 and 4), video ethnography and ethnomethodological video-analysis (Paper 1 and 3), and a design workshop (Paper 3). Chapter 2 provides an overview and description of specific methods of data collection and data analysis used in the papers.

Each of the papers makes a separate contribution that goes beyond the focus of this kappa, but together they help to advance the studies of ethics in practice in HCI and CSCW with the empirical research on ethical responsibility, tech workers feelings and agency, and ethical reasoning in tech workplaces. The main findings from the studies and their synthesis in relation to the three research questions are presented in Chapter 3.

1.4 The scope

The main purpose of conducting empirical research is to develop a theoretical argument around the research questions drafted above. However, one natural limitation comes from the groups that I studied. The tech workers in this

research project are mostly based in Europe and come from a diverse set of industries and businesses. Since I used mostly qualitative methods and did not attempt to create a representative sample of tech workers, I am not aiming to make statistically representative findings. Instead, I am interested in making a theoretical argument about the types of situated ethical reasoning in tech. Some of these theoretical arguments can hopefully be advanced in the future studies and, perhaps, be rephrased into hypotheses that can be checked in the quantitative studies.

In this work I chose to not take a post-human approach to ethics, which means that I focus on people and communities instead of looking at technologies or institutions as actors – this is a limitation, but also a choice that fits my research interests. I do see a problem of incompatibility between an individualistic take on responsibility and the complexity of technology developments, and I am interested in how tech workers approach this problem.

1.5 Contributions

With my research, especially papers 2 and 4, I hope to bring the academic studies of tech ethics a little closer to the folk approach towards ethics and morality accepted by tech workers. Understanding how the tech workers, who do not work with ethics as their main occupation, approach their work can be useful for developing educational programs for engineers and designers, as well as guidelines, theories and recommendations better fitted for those who work in the industry. One important finding I made was that tech workers do care about ethics, but often limit their ethical responsibility to the moment of choosing a job and taking care of their immediate work environments.

In all of the papers, I demonstrate the limitations of individualist ethical reasoning and the need for developing structures for collective action. However, papers 2 and 4 indicate that the potential for any kind of collective action, bottom-up organisation, or even for forming a shared agenda around ethics may be rather limited among the workers I studied. Even the collective action in defence of tech workers themselves was rather limited and not often successful, when it comes to preserving tech workers benefits.

One of the key contributions of this thesis is bringing affect and emotion into the conversation about ethics of tech and pointing to the shared nature of moral affects[3]. There is a growing amount of research stressing the importance of

[3]There is a discussion about the use of concepts of affect and emotion (Åhäll 2018; Hemmings 2005), but for the purposes of this work I use them interchangeably. In that, I align with the authors who argue against maintaining epistemological dualism between bodily and cultural/political. When the distinction between affect and emotion is made, affect usually refers to the bodily reaction, something subconscious and streeped out of meaning such as the body registering some physical sensation. Emotion refers to the process of making sense out of affect with the help of cultural resources. Such dualism reinforces the distinction between body and culture (and often the hierarchy between the two) – something that can be theoretically

affect for tech ethics (Amrute 2019; Su, Lazar, and Irani 2021), yet a more established approach to ethics still relies on the distinction between rational and emotional, where ethics can only be discussed through the language of values and ethical principles. I look at emotions and affect as individual experiences that share some structural similarities (Ahmed 2004). My intermediate conclusion is that moral discomfort around ethical issues may be more common for those occupying lower status positions within the tech sector (more about that in Paper 2 and 4). In all of the studies the mentioning of emotions serves as an important indicator of the ethical issue at hand, but also the problems with power and responsibility distribution. Negative emotions most often marked situations of power imbalance and the resulting impossibility to act in accordance with one's values. I looked at emotions discursively, studying how tech workers talk about their feelings in the interviews, rather than measuring them or looking at non-verbal expressions of emotions.

To sum up, I investigate ethical reasoning of tech workers in situ through a combination of interviews, participant observations, and video recordings of workshop activities. By focusing on tech workers and their situated reasoning, I also hope to strengthen an anthropological perspective on ethics in fields of HCI and CSCW, and explore ethics in the context of everyday work and everyday power structures within the tech sector[4].

1.6 The list of the papers and my contribution

1.6.1 Vulnerability as an ethical stance in soma design processes.

Kristina Popova, Rachael Garrett, Claudia Núñez-Pacheco, Airi Lampinen, and Kristina Höök. 2022. In Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 178, 1–13.

We articulate vulnerability as an ethical stance in somadesign processes and discuss the conditions of its emergence. We argue that purposeful vulnerability – an act of taking risk, exposing oneself, and resigning part of one's autonomy – is a necessary although often neglected part of design, and specifically soma design, which builds on felt experience and stimulates designers to engage with the non-habitual by challenging norms, habitual movements, and social interactions. With the help of ethnography, video analysis, and microphenomenological interviews, we document an early design exploration around drones, describing how vulnerability is accomplished in collaboration between

problematic (Åhäll 2018).

[4]They may be important for broader conversations about ethics in sustainability domains, for example, since the nature of the problem there is very similar – how to approach individual responsibility in the context of a complex social problems.

members of the design team and the design materials. We (1) define vulnerability as an active ethical stance; (2) make vulnerability visible as a necessary but often neglected part of an exploratory design process; and (3) discuss the conditions of its emergence, demonstrating the importance of deliberating ethics within the design process.

I contributed to the paper the conception, data collection (ethnographic notes, video data) and data analysis (video, ethnographic notes, micro phenomenological interviews), writing most of the sections (with the help of all contributors), editing and bringing together contribution of other authors, responding to the reviewers comments and editing the paper with the help of my advisors.

1.6.2 Who should act? Distancing and vulnerability in Technology Practitioners' Accounts of Ethical Responsibility.

Kristina Popova, Clàudia Figueras, Kristina Höök, and Airi Lampinen. 2024. Proc. ACM Hum.-Comput. Interact. 8, CSCW1, Article 157 (April 2024), 27 pages.

Attending to emotion can shed light on why recognizing an ethical issue and taking responsibility for it can be so demanding. To examine emotions related to taking or not taking responsibility for ethical action, we conducted a semi-structured interview study with 23 individuals working in interaction design and developing AI systems in Scandinavian countries. Through a thematic analysis of how participants attribute ethical responsibility, we identify three ethical stances, that is, discursive approaches to answering the question 'who should act': an individualised I-stance ("the responsibility is mine"), a collective we-stance ("the responsibility is ours"), and a distanced they-stance ("the responsibility is someone else's"). Further, we introduce the concepts of distancing and vulnerability to analyze the emotion work that these three ethical stances place on technology practitioners in situations of low- and high-scale technology development, where they have more or less control over the outcomes of their work. We show how the we- and they-stances let technology practitioners distance themselves from the results of their activity, while the I-stance makes them more vulnerable to emotional and material risks. By illustrating the emotional dimensions involved in recognizing ethical issues and embracing responsibility, our study contributes to the field of Ethics in Practice. We argue that emotions play a pivotal role in technology practitioners' decision-making process, influencing their choices to either take action or refrain from doing so.

I was responsible for the general conception of the paper that was established in dialogue with Claudia Figueras, introducing relevant theoretical frameworks, and conducting half of the empirical data collection. Claudia Figueras has done the interviews with government sector workers in Sweden, I did the interviews with people in academia and some industry professionals. I led data analysis, which was conducted together with Claudia and was responsible for writing

most of the text with a significant contribution of Claudia, as well as editing and responding to the reviewer's comments together with other co-workers.

1.6.3 Voicing Discomfort, Communicating Breakdowns: Design Exploration of Felt Ethics in a Body-Based Workshop.

Kristina Popova, Arife Dila Demir, Joo Young Park, and Rob Comber In Nordic Conference on Human-Computer Interaction (NordiCHI 2024), October 13–16, 2024, Uppsala, Sweden. ACM, New York, NY, USA, 13 pages.

This paper reports on a body-based exploration of felt ethics through a workshop with interaction designers. Contributing to developments of hands-on methods to approach ethics, we engaged with ethics through the experience of discomfort from interpersonal interaction. Our workshop methodology is grounded in soma design approach, while the theoretical roots of our work are in the ethics of care. In this paper, we analyse the workshop results based on video- recordings and post-workshop interviews with participants. Using a qualitative inductive approach, we consider (1) what constitutes discomfort in the interaction between workshop participants, and (2) how participants make this discomfort shared. Through that, we identify the ways in which bodies feel and display discomfort, and how it extends our understanding of ethics. We reflect over the potential, limitations, and risks of body-based workshops to explore design ethics, i.e. the norms and values of design practice, as a felt phenomenon.

Together with Dila Demir, I was responsible for organising and conducting the workshop and collecting data. I conducted most of the data analysis, generating the main idea and doing most of the writing of the paper, as well as coordinating the process.

1.6.4 Making the World a Better Place? Ideology, Discipline and Agency in The Tech Sector

Kristina Popova. Unpublished manuscript

How do technology practitioners explain to themselves and others why what they do is ethical? And how free are they to "do good"? IT and engineering have a long history of thinking of themselves as problem solving disciplines helping the world to become a better place. Although such a positive outlook has recently been challenged, the main narrative of problem solving persists. How much freedom to change things within their companies do they have? The paper answers these two questions based on 20 in-depth interviews with tech practitioners working in industry. We contribute to studies of ethics in practice by demonstrating that tech workers make their main ethical choice when accepting a job offer. The internal culture of tech companies and the per-

sistence of the problem-solving narrative restrict collective action and limits the potential for change from within.

The paper is a single author project, where I was responsible for writing and data collection. The discussions over data were happening together with my supervisors Rob Comber and Airi Lampinen.

2 What is Ethics?

'Ethics' is a contested terrain. Working at a university, I often encounter the word ethics in the context of 'we need to solve ethics', which means that we need to get a formal ethical approval from the regulating bodies to get the research started. This is probably not the most common understanding of ethics outside the university, yet bureaucratic practices of getting ethical approval, collecting informed consent, or complying with GDPR are associated with ethics as well as philosophical theories of virtue ethics and utilitarianism, or discussions on justice, fairness, and care. A conventional understanding of ethics as 'compliance' (Orr and Davis 2020) is strong both within and outside university and in relation to technology. That is why ethics is contested – there is a discursive conflict between understanding ethics as a matter of compliance, politics, or self-development.

Ethics also relates to the question of what it means to live a good life or what is it that makes life bearable (Lloyd 2015), ethical codes of conduct that constitute the norms of professional behaviour, such as the ACM Code of Ethics and Professional Conduct – these are just a few examples of ethics in academic discourse. A folk understanding of ethics is about behaviour in difficult situations, solving the ethical dilemmas, when moral norms clash and conflict with each other. These understandings do not necessarily overlap.

Besides ethics, there is also a concept of morality. In the History of sexuality Foucault distinguishes between the two meanings of morality: morality as a set of values and rules of action that are recommended to individuals, and behaviour of those individuals (Foucault et al. 1978). The distinction is similar to the one between plans and situated actions (Suchman 1987), ethical guidelines and the situated work of practitioners explored by the studies of ethics in practice (Lindberg, Karlström, and Männikkö Barbutiu 2021; Widder, Zhen, et al.

2023; Wong, Madaio, and Merrill 2023).

Ethics is separate from morality – it is a philosophical theory of morality or a practice through which one becomes a moral subject (Foucault 1997). For Foucault, ethics is a practice of subjectification, the process of becoming moral by building certain relationships with the self – becoming an ethical subject through disciplining of the self (Foucault 1997; Foucault et al. 1978).

> Ethics is a deliberate practice through which an individual forms themselves as a subject, in relation to prescriptive norms, through acts of self-care, self-discipline, and self-transformation (Foucault 1997)

Ethics is about the internal regulations, whether they are rules of reasoning (Rawls 2005), sentiment (Noddings 2013), or practice (Foucault 1997) that allow people living together in a society worth living. There are debates in moral philosophy on what ensures this living together, living the life worth living (Lloyd 2015) in a society worth living. Since this is not a thesis in philosophy, my presentation of philosophical theories on ethics will be rather fragmented. This chapter will discuss some of the approaches to ethics that are relevant in the context of HCI and tech work, where researchers are working on making ethics more concrete and approachable in the context of tech development.

2.1 Ethics of technology and ethics of technology practitioners

Science and technology studies consider technology to be a mediator that conveys values of its creators (Winner 1980). Technology does contain the past in the present in amplified ways, and in that sense incorporates ethics and helps to reaffirm certain ethics in favour of others.

Technologies do convey values and presume various affordances, 'programs of actions' (Latour 1992). A program of action is a rule or a set of rules embedded in an artefact. Technology has a possibility to convey more than one program at the same time, for example, a seat belt can hold one tight but also let the driver change the gears without giving him much trouble. Those programs of action can have political implications and reasons of implementation (Winner 1980), i.e. they can be related to the interests of different social groups. However, technology is not deterministic but 'multicited' (Ihde 1990) and completed in the process of its use and interaction with other technology.

The agency of technology implies that ethics is embedded in design, hence, design practice is always normative, shaping "how things ought or should be", reinforcing certain ways of being in the society (DiSalvo 2012, s.16). In this thesis, I operate from the assumption that technology embeds values and, to a degree, the conditions of its production.

There is no singular understanding of 'ethics' in HCI. There is a long tradition of reflecting upon research ethics in HCI (Acker 2006), not harming or deceiving

research participants and distributing power in the design process in fair ways (Spiel et al. 2018). The recent wave of ethics discussions concerns AI and its consequences – that is the field of AI ethics with its own agenda . The spread of ML-based technology laid ground for the discussions on bias and fairness, transparency, sustainability in AI systems – and a more critical discussion on structure of power and the development of AI, stating that discourse of AI ethics replaces a much needed conversation about power and the need for political regulation of tech corporations (Farrell and Fourcade 2023). [1] At the same time, other ethical issues relevant for the discipline go beyond one particular technology and include the questions of design values (Friedman and Hendry 2019), equality and justice (Dombrowski, Harmon, and Fox 2016), power redistribution in design process, working with marginalised groups and with 'sensitive', taboo, topics.

My initial approach to ethics was radically empirical. Attempting to avoid using one definition of ethics, I was focusing on the participants' use of the category of 'ethics' instead. By doing so, I tried to explore ethics as it was understood and practiced by tech workers. In the first two papers, I was focused on ethical practices, including the discursive practices of defining certain areas of work as 'ethical'. Approaching ethics 'in situ', I presumed that ethical values can be studied in the process of design and creation of technology rather than only after technology has been released in the world and met the end user. This research exploration was situated, broadly, in the tradition of studies of ethics in practice in CSCW (Boyd and Shilton 2021; Chivukula et al. 2020; Lindberg, Karlström, and Männikkö Barbutiu 2021).

While ethical guidelines are useful for operationalising complex ethical norms, the studies ethics in practice claimed that understanding ethics 'in the real world' of tech development requires paying attention to practical application of norms and the organisational contexts of workplaces where ethical guidelines are implemented (Madaio et al. 2020). The studies of ethics in practice challenge the attempts to standardise ethics as a matter of universal rules and guidelines. Instead, the scholars of this direction aim to understand the work of technology production as situated in the specific work practices and organisational contexts (Wong, Madaio, and Merrill 2023).

This line of study has been focused on the practices of tech workers of implementing those rules and regulations in practice or trying to act in line with their ethical norms in specific contexts. The two important insights from this line of study is that (1) rules are always implemented by concrete people whose actions are situated within the realms of concrete institutional domains (2) placing too much blame on an individual designer or an engineer can be a way to avoid

[1] This thesis is not dealing with the questions of AI ethics as such, but my study is influenced by the studies of ethics in practice in AI domain (Figueras, Verhagen, and Cerratto Pargman 2022) and Paper 2 talks about responsibility and emotion in development of complex AI systems.

the conversation about power, since developing individual reflexivity without giving power to implement the action will not lead to a positive outcome, yet may have high cost for the involved individuals (Powell et al. 2022; Widder and Nafus 2023).

The related approaches to ethics in HCI include the ethnography of value-work in tech sector done by Shilton (Shilton 2013), micro-ethics by Spiel (Spiel et al. 2018) or in-action ethics by Frauenberger (Frauenberger, Rauhala, and Fitzpatrick 2016). The scholars of ethics in practice have recently started to pay attention to the emotional component of 'ethics' – dealing with critique (Su, Lazar, and Irani 2021) or making decisions to take or avoid taking responsibility for ethical action (Widder and Nafus 2023). My work contributes to the studies of ethics in practice with empirical studies of ethics in first person design, specifically focusing on emotions as one of the most important domains for first person design.

2.2 Ethics of Care

My work is influenced by the ethics of care theorists, who argued in favour of studying ethics in the concrete situations and not in the abstract principles (Tronto 1993). As defined by Carol Gilligan and summarised by Joan Tronto, there are three principles distinguishing ethics of care:

- Focus on responsibilities instead of rights;
- Focus on circumstances rather than formal and abstract rules;
- Focus on the activity instead of a set of principles (Tronto 1993, pp. 79).

As highlighted in the quote below, Ethics of Care differs from the previous theories because it suggests to focus on the details of specific situations rather than universal guidelines and the acts of morality instead of moral norms. What is important when it comes to defying care is that care, as an everyday moral practice, is work – it is not an intention, but a specific activity to 'maintain, continue, and repair our world so we could live there as well as possible" (Tronto 1993, pp. 103), and it is an activity that is directed to others and is based on recognising the needs of the other.

> "First, the ethic of care revolves around different moral concepts than Kohlberg's ethic of justice, that is, responsibility and relationships rather than rights and rules. Second, this morality is tied to concrete circumstances rather than being formal and abstract. Third, this morality is best expressed not as a set of principles but as an activity, the "activity of care." In Gilligan's different voice, morality is not grounded in universal, abstract principles but in the daily experiences and moral problems of real people in their everyday lives" (Tronto 1993, pp. 79).

The theorists of care ethics were also first to bring emotions in the conversation about ethics – something that is very important for the consideration of ethics in the first person design. Nodding developed an argument that acts of morality are grounded in feeling. According to her, to care for means having a feeling and a natural desire to act in favour of the other. Nodding distinguishes between the 'natural caring' and 'ethical caring' –the first one is done out of love, the second must be done out of reason (Noddings 1984). In contrast to classical philosophical studies of morality that were focused on rational grounds of behaving ethically, Nodding grounds morality in the felt experience.

> One either feels a sort of pain in response to the pain of others, or one does not feel it. If he does feel it, he does not need to be told that causing pain is wrong. If he does not feel it in a particular case, he may remember the feeling—as one remembers the sweetness of love on hearing a certain piece of music—and allow himself to be moved by this remembrance of feeling. For one who feels nothing, directly or by remembrance, we must prescribe re-education or exile. Thus, at the foundation of moral behavior—as we have already pointed out—is feeling or sentiment. (Noddings 2013, pp. 86)

Nodding's approach to emotion and care, specifically her take on care as a natural and inherently feminine practice, was criticised for de-evaluating the labour of care and for disregarding the political and economic dimensions of care. Tronto and Federici point out that care work is usually underpaid or unpaid and most often carried out by people who belong to low social and economic classes. Disregarding the political dimension of care is beneficial to the status quo and benefiting those women who already are powerful and privileged (Tronto 1993, pp. 91). As Michele Murphy puts it:

> "Projects of care, feminism and otherwise are full of romantic temptations that disconnect acts that feel good from their geopolitical implications." (Murphy 2015, pp. 175)

At the same time, despite the critique, Marxist and postcolonial feminist authors do acknowledge the emotional dimension of care. They note that emotional work is the part that most often gets devalued and excluded from being considered work. They also point at the importance of political context of feeling (Ahmed 2014).

2.3 Emotion and affect in feminist epistemology

The importance of attending to felt in feminist theory is related to the situated nature of knowledge, which implies that the ability to know depends on the social position of the subject of knowledge (Haraway 2020; Toole 2019) and the phenomenon of epistemic oppression. Epistemic oppression refers to the absence of underprivileged groups that have embodied knowledge of injustice

from knowledge production. The paradoxical situation is that the people with experience of marginalisation have both better access to knowledge (epistemic privilege) of injustices and oppressions than those in power and lack resources to voice and legitimise their concerns, because they have been excluded from the process of knowledge generation and meaning-making. Epistemic injustice relates to the fact that those who have been excluded from knowledge production lack legitimised tools to express this knowledge:

> Epistemic agents are marginalised when they are excluded from meaning generating, or interpretive practices. A meaning-generating practice is one that guides, shapes, or governs our way of thinking about things (Toole 2019, pp. 604)

This lack of conceptual resources and underrepresentation in knowledge production, marks the importance of attending to the felt. Even though the knowledge may not have been articulated, oppression is registered in the body. Injustice is registered through experience; the experience generates the need to make sense of them. This is where feminist epistemology is needed: 'Feminism is a sensible reaction to the injustices of the world, which we might register at first through our own experiences. We might work over, mull over, these experiences; we might keep coming back to them because they do not make sense. In other words, we have to make sense of what does not make sense.' (Ahmed 2017, pp. 21).

Experiences have been pushed away from the conversation about ethics in moral philosophy and from the traditional conversation about ethics focused on the 'rational' grounds of taking or not acting. 'Emotions are associated with women, who are represented as 'closer' to nature, ruled by appetite, and less able to transcend the body through thought, will and judgement' (Ahmed 2004, pp. 3).

An alternative take on emotions is to consider emotions as distinguishing the boundary between the inside and the outside of a subject not as "something I or we have", but something that creates a boundary between Inside and Outside (Ahmed 2014, s.10). Emotions are registered in the body as a reaction to the events outside the body, and experiences are the first reaction to registering injustices of the outside world. They mark the intense contact between the body that is not at ease with the world and the world. Noticing and registering the sensations is then the first step in recognizing the nature of the encounter with the world, the second step would be offering new ways of understanding the world (Ahmed 2017; Hemmings 2005; Hemmings 2012; Highmore 2016; Åhäll 2018).

3 Methodology

"Our studies would be more convenient if we could begin with a definite idea of where a study would lead, employ standard research methods, and end up pretty much where we thought we were going. However, if research worked this way, the world would be pretty dull; there'd be little reason for doing it."
(Livingston 2008, pp. 177)

3.1 EMCA as practical philosophy of ordinary action

This thesis was written within an interdisciplinary field of HCI. The papers combine several methods and experiment with mixed-methodological approaches, but the initial intention to approach ethics by observing everyday work stems from ethnomethodological tradition of respecifying theoretical phenomena as matters of practice (Garfinkel 1967). In this chapter, I will attempt to sort out and generalise on the methodology that connects the different papers collected in this thesis and on the knowledge-generation approach used in this thesis.

Livingston calls ethnomethodology a practical philosophy of 'ordinary action' (Livingston 2008). Ithnomethodologists studied the most mundane of human activities, such as making a puzzle or having a phone conversation (Sacks 1992), to talk about reasoning (Livingston 2008), cognition (Goodwin 2000), knowledge (Lynch 1993), social order (Garfinkel 2002), and other theoretical concepts. This approach is vastly different from conventional sociology, the home discipline of ethnomethodology. So is the ethnomethodological tradition of respecifying theoretical phenomena as matters of practice (Garfinkel 2002).

Respecification implies that, instead of using theoretical constructs to explain phenomena, ethnomethodologists study members' methods for producing these

Table 3.1.1: Combinations of methods used in the studies based on my fieldwork.

	Methods
Paper	Methods Used
1	participant observations + video analysis
2	qualitative interviews
3	workshop + qualitative interviews + video analysis
4	qualitative interviews

phenomena. Some examples include studies of doing gender (Garfinkel 1967), doing science (Lynch 1993; Sormani 2014), and even reasoning (Livingston 2008). Ethnomethodologists studied how sociological categories and even cognitive phenomena (vision) (Livingston 2008; Goodwin 2000) are created through coordinated observable actions of people, instead of taking these categories as explanations of why people behave in a certain way (Garfinkel 1967). By doing so, *ethnomethodology challenges the hierarchy of knowledge between a social scientist/analyst/academic researcher and a competent practitioner, a member of society being studied.*

Radically, from the perspective of a more traditional sociology, ethnomethodology assumed that both an academic social science researcher and a competent member of society have equally valuable knowledge of the phenomena of social order. Competent members of society are also the analysts, the researchers, because it is their ability to recognise and reproduce the social phenomena that creates social order. An analyst, a researcher, on their turn has to be a competent member of the society to produce sound accounts of members' practices (Garfinkel 1967). The difference between researchers and non-researchers then mostly resides in the fact that not all competent members of society are also competent sociologists and/or academic researchers capable of generating academic accounts of social phenomena.

Early ethnomethodology goes against the hierarchy of knowledge between the 'lay members of society' and academic researchers. I followed this logic at the beginning of my study, which I started from trying to understand how my research participants understand ethics, i.e. what is the folk understanding of ethics, and which moments do the research participants consider as ethically challenging. However, since I rely on 'non-members' analytical categories in the papers and in the discussion of this kappa, this research is inspired by ethnomethodology but does not strictly adhere to ethnomethodological tradition.

3.2 Data collection

I used a variety of qualitative data collection methods, including participant observation, interviews, and video recordings. Paper 1 relies on video recordings, participant observations and micro-phenomenological interviews. Papers 2 and 4 are based on semi-structured interviews; Paper 3 is based on the video recordings of a body-based design workshop and semi-structured interviews.

3.2.1 Participant observations

Participant observations supported by notes and video recordings serve as the basis for Paper 1. I had mainly collected the observations during my first year as a PhD student, when I participated in a research-through-design (Zimmerman, Forlizzi, and Evenson 2007) process with four other design researchers, co-authors of Paper 1. During the ideation stage of the design work, I took written notes and video-recorded the activities to analyse them later. Being involved in these activities as a full-fledged participant, I also approached them as an ethnographer and observer, trying to understand the meaning of the activities and to see how the values of this community are translated into and manifest in the designs and the practices of working together. The results of these explorations are presented in the first paper.

3.2.2 Video Analysis

Video recordings serve as the primary data for papers 1 and 3. In both cases, I used video to document the interactions among workshop participants. I used static cameras to capture as many details as possible. For Paper 1, the video recordings of a drone crash were approached through EMCA video-analysis (Heath, Hindmarsh, and Luff 2010), which was later transformed into ethnographic vignettes, and video recordings of the moments of different degrees of discomfort that constituted a normal practice during those workshops.

For Paper 4, the analysis was conducted through a combination of inductive categorisation of video and more in-depth video analysis of selected shorter video fragments. In general, the procedure for analysing video included several steps:

- reviewing the videos,
- coding videos according to my research questions,
- grouping the fragments related to the same theme,
- transcribing or annotating selected fragments,
- performing a detailed video-analysis of selected fragments,

- conducting data sessions to analyse the videos together with other researchers.

3.2.3 Qualitative Interviews

Two types of interviews were used in the papers: semi-structured interviews for papers 2 and 4, and micro-phenomenological interviews for paper 1.

Micro-phenomenological interviews focused on the details of the experience, the participants' memories of breakdown and repair leading up to the drone crash. All questions derived were not planned in advance. We also used video recordings to prompt participants' memories. The idea behind combining these methods with video analysis was to combine the "first" and "third" person perspectives on the event.

The qualitative semi-structured interviews provided data for Papers 2 and 4. For the Paper 2, I conducted 10 semi-structured interviews with interaction design researchers, mostly working in academia. The idea behind the interviews was to explore how participants understand ethics and how they define it in their work. Hence, I did not provide the participants with any definition of ethics, but included open-ended questions such as "Have you ever encountered any ethical issues in your projects?" and "Have you encountered any ethical issues during planning/implementing/presenting your project?" Major parts of each interview were centred on follow-up questions prompted by participants' initial answers, engaging with the details they viewed as ethically challenging. The interviews centred on the details of the participants' design work and the aspects of their work they perceived as ethically challenging. This data set was broadened by 4 additional interviews after conducting the first round of analysis.

As a second step in the project, I compared interviews with interaction design practitioners with interviews with AI developers working in Swedish governmental organisations. The second set of interviews was conducted by Claudia Figueras, a PhD student at Stockholm University (at that time). The two of us conducted a preliminary analysis of a joint data set, which led us to conclude that the two groups of technology practitioners had very different approaches to taking or avoiding responsibility for ethical action.[1]

For the project "Exploring discomfort" (Paper 3), together with Demir as the second author we conducted post-workshop interviews with workshop participants. The initial questions prompted the interviewees to share their experience of participating in different workshop exercises and their recollection of the moments of discomfort experienced in the workshop. Similar to micro-phenomenological interviews, our questions aimed to elicit participants' felt

[1]The analysis focused on tech practitioners' discourses, but I consider language to be part of practice (Potter 2003) and assume the interplay among language, action, and affect.

experiences and, to a large extent, built on the information they had shared earlier in the interview.

Paper 4 is based on in-depth interviews with tech workers at commercial companies in Europe. The roles held by the interviewees, current or previously, included software engineers, technical writers, project managers, and UX designers. They are all relatively young (most were under 35) and in the early or mid-career (3 to 10 years of experience, besides one interviewee). The interview scripts were rather broad and focused on several aspects of the practitioners' work in the company, including their job tasks, emotions at work, relationships with colleagues, and attitudes to responsibility and ethics. The interviews had a flexible structure, with follow-up questions, the priority of guiding the conversation was given to participants.

3.2.4 Body-based workshops

Workshops are a common form of knowledge production in design research, ideation and evaluation. In the papers 1 and 3, I relied on workshops for data collection (generating video data), but also for discussions, creating networks between other design researchers and probing research agendas (Chicau, Popova, and Fiebrink 2024; Popova, Chicau, et al. 2024; Popova, Park, and Demir 2023).

With collaborators for Paper 3, I conducted a series of movement-based workshops with elements of contact improvisation to encourage understanding: how does discomfort manifest itself in the interaction, when it emerges and which work do the participants do to hide the discomfort. The list of the workshops is presented below:

- 'Making Ethics Strange' – Workshop in April 2022 with WASP HS Researchers.
- A workshop with the students from Soma design course at KTH, May 2022.
- The Workshop on using pain as a design resource at Nordes 2023 with Dila Demir and Jooyoung Park (Popova, Park, and Demir 2023).
- From Individual Discomfort to Collective Solidarity: Choreographic Exploration of Extractivist Technology TEI and ECCSW workshops (Popova, Chicau, et al. 2024; Chicau, Popova, and Fiebrink 2024).

The workshops usually included from 5 to 12 participants, most of whom either worked in interaction design or HCI or were students in interaction design. We started from a short introduction where the organisers explained the workshop's purpose. Then the participants would fill in body maps and share the way they were feeling with other participants. After that we would have one or several body-based exercises, usually in a pair. Then they would discuss this

experience in pairs, fill in another round of body maps, and share their experiences in the big group. In some workshops we would have several exercises and the following sharing sessions.

The Workshop on using pain as a design resource at Nordes (Popova, Park, and Demir 2023) also included an ideation stage, where participants used design materials to express their discomfort. During this workshop we were interested in the uncomfortable beyond the room, where the workshops were conducted. The participants' chose to design on the experience of discomfort in their studies, for instance, on working with pain in medical settings.

3.3 Data analysis

My approach to data analysis combined inductive and theory-driven techniques. Some common principles that I applied in my thesis were:

- pragmatism and empiricism – approaching theoretical categories as practical phenomena;
- data-driven theorising – combining inductive qualitative analysis with conceptual generalizations;
- methodological diversity – shifting between video, participant observations, interviews and different ways of presenting the findings from those.

Overall, my approach to data and theory in these works combines a theory-driven approach with empirical analysis. I did have a certain theoretical interest and hypothesis before approaching the data, but I relied on the data to generate and refine theoretical concepts. For instance, the idea of finding ethics in the moments of breakdowns and discomfort was a combination of empirical observations (Paper 1, Paper 2) with readings in moral anthropology (Zigon 2007), while the theoretical stances came directly from the comparison between the data sets for Paper 2.

In Paper 1, I experimented with a combination of video-ethnography, EMCA-based video analysis, and micro-phenomenological interviews, analysed with the help of thematic analysis. As the first step, I created the detailed EMCA analysis of the transcript of the video recording of a drone crash and the repair work conducted by the group members, using a modified Jefferson transcript (Jefferson 2004).

The transcript allowed us to see the repair work the group members performed after the accident to continue the group ideation process. In the publication, I chose to present the transcript as an ethnographic vignette, since unfamiliar readers find transcripts difficult to understand.

Paper 2 relied on the inductive comparative analysis of the two interview sets. The research was conducted in several steps. At the first step, I analysed interviews with interaction designers. The purpose of this data was to understand

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43    R           Are you okay, did it hurt?
44                (1.55)
45    C           a little bit here but I don't think it was anything
46                (1.62)
47                it's on the chin
48                (0.98)
49    R           Sorry, I need to put that down
50    C           that's okay\
51                (0.74)
52    R           I am so sorry! [I was
53    C                                    [oh! it's a propeller here
54                (0.11)
55    K           laug[hter
56    C                 [hahahaha
57    R                 [hahahaha
58    C           that's fine
59                (0.34)
60    K           It's a very (.) rich day today
61                (0.35)
62    R           Sorry
63                (0.03)
64    C           that's okay
65    R           I am thinking you might have a knot [in your hair
66    C
                                                     [yeah no no no no worries
67    C           that's fine
68    R           shiiiiit=
69    C           =it's nothing, it's a little accident
70                (1.25)
71    C           I think
72                (0.04)
73    R           I thought I was making it go the opposite way
74                (0.05)
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Figure 3.3.1: Transcript A detailed transcript from the drone crash episode that was later transformed into an ethnographic vignette.

pragmatically how researchers approach ethics in their work. The interviewee described a variety of situations as 'ethical', but the common denominator was that they were emotionally charged.

I used the categories derived from this analysis to approach the data in the second data set. Based on inductive qualitative analysis of the combined data set, we have identified a set of theoretical categories that can be used for classifying the data – the three ethical stances, which present the main theoretical contribution of this paper.

Paper 3 combines theoretically informed interest in the experience of discomfort with inductive analysis of video data from a workshop. In this paper, I experimented with a variety of techniques for analysing video data and presenting results. At first, I coded the video data using categories of discomfort, negotiation, and laughter[2]. I then edited videos to create longer sequences, combining moments of laughter, expressions of discomfort (when the word 'discomfort' was used), and participants' comments on the uncomfortable interactions in the workshops. My goal was to use video snippets instead of transcripts to present and analyse the data. I have also conducted a similar inductive analysis to compare how participants talk about their interactions in the interviews with those in the group sessions.

[2] Laughter was identified as one possible marker of discomfort in Paper 1.

In Paper 4, the analysis was done through inductive coding of the interviews in Atlas.ti. I did a few rounds of the analysis; the first round was conducted inductively, then four main groups of codes were developed: communication, conflict, emotions, and organisations. Several interviews were coded independently by my supervisor and me to establish the relevance of the codes. After half of the data had been analysed, I ran a few collaborative discussions with the supervisors to refine the themes. The established codes were then applied to the remaining data.

3.4 Research ethics

This research project did not require ethical permission from the university or the national ethical board. All the workshop participants and interviewees have given their informed consent to participate in the study or agreed to generate the data. I have anonymised the data, and no real names of the participants are mentioned in the publications or the kappa. Because of anonymity, the descriptions of professions and other characteristics that could de-anonymise participants have been changed or omitted. I have separately consulted the research participants regarding specifically sensitive data and whether they were okay with the quotes being included in publications. The data were stored in a safe space and were not shared outside the research group. The transcripts are made with the KTH transcription service.

Additionally, during the workshops we ensured that the levels of discomfort stays low, we tried creating a relaxed atmosphere and voiced that participants should not be continuing if they do not feel comfortable with some of the workshop activities (however, as I claim in paper 4 this is not really enough to say 'you can always withdraw'). In short, I tried to ensure that data collection and participation in the research was consensual, did not harm the participants, and that the final representation of the data was true to participants stories.

4 Results

This section summarises findings from the four papers in this thesis. I will describe the contributions of each paper and summarise the papers with a focus on agency, emotion, and responsibility. As mentioned before, the papers come from different research projects with different research logic: Papers 1 and 3 take a lot from Design Research, while Papers 2 and 4 have a sociological orientation.

Each of the papers takes up different aspects of ethical reasoning in the context of tech workplaces, which include academic HCI research, the governmental sector and commercial companies. I focus on communicative practices and discourses around ethics at these workplaces to study ethical reasoning empirically as an observable phenomenon and a part of everyday work.

4.1 Paper 1. Ethics, Vulnerability and Breakdowns: what is taken for granted.

The first paper sets up the scene for future exploration of ethics at tech workplaces. The focus is on my workplace at the university and the activity of a research group engaged in design ideation. While the environment of an HCI department at the university is different from a software development company or a startup, it is also a tech workplace, where prototypes of technology are designed, ideas are developed and, especially for the AI sector, technologies are created.

Within the realms of the PhD, the goal of the paper was to work out a methodology of studying ethics as a phenomenon observable in face-to-face interaction at workplaces. I started from an assumption that, while ethics is often understood as a matter of guidelines and theoretical principles, it is also a practical achievement and a matter of the everyday work of tech practitioners. But what does it

mean to approach ethics empirically in tech work? What should we look for? Paper 1 attempted to provide preliminary answers to these questions.

Paper 1 suggested looking at the moments of **breakdowns** in the everyday work of tech practitioners as a way to analyse ethics in practice. The concept of breakdown is often used in sociology – breakdowns, as the moments when taken for granted norms are no longer applicable, have long been interesting for social scientists (Garfinkel 1967; Becker 2018). In moral anthropology breakdown represents the moments when taken-for-granted behavior is no longer possible and ethical decision making takes place (Zigon 2007). A similar idea was used in early EMCA experiments (Garfinkel 1967) – when the norms of the everyday are disrupted, they become observable for analysis [1].

The breakdown discussed in the paper was a literal breakdown, an episode when we crashed a toy drone during our design exploration. This episode represents a disruption of the routine, when the group members have to stop ideation and get involved in the repair process (Schegloff, Jefferson, and Sacks 1977). What needs to be repaired is the group cohesion, which has been disrupted when one group member accidentally crashes the toy drone into another. This was a mild incident without any long-lasting consequences, yet the repair work done by group members can be illustrative for our studies of ethics in practice.

In the paper, we analyse the **interaction work** that the research group performs to fix the aftermath of a drone crash – making sure that no one was hurt, there is no resentment in the design group. This interactional work is not seen as a key part of tech work, at least it is not formally reported as such, but it is important and highlights the values that the participants care about. In this case, the peace and well-being of the ideation participants.

As the main contribution of the paper, we **illustrate that ethics is not only a matter of institutional rules; ethics is situated in the everyday interactions, bodies and behaviours**. Ethics in mundane becomes specifically visible during breakdowns, when tech workers have to do the work to preserve the status quo. Breakdowns are the moments when ethics becomes visible and open for empirical exploration. A conclusion to be made is that it is possible to study ethics as part of design practice, through the work that designers perform for reestablishing certain values. In that case it was normalising vulnerability and avoiding conflict, in this case. Since we do not advance towards actually creating technology, here I can only talk about the ethics of design work rather than the ethics of technology.

Another central thing in the paper was articulating **vulnerability** as a value in a first-person approach – the design framework within which we have con-

[1] Later ethnomethodology went away from the idea of breakdown, preferring to study social order in the naturally occurring everyday situations rather than artificially created breakdowns (Garfinkel 2002)

ducted the ideation process. We defined vulnerability as a deliberate ethical stance taken by a first-person design practitioner, who views their own vulnerability as a ground for design; and not only acknowledges the presence of their subjective experiences, but considers them as justification for design decisions. Vulnerability is not among the 'members categories' (Sacks 1992) – not anything we used during design sessions, yet we argue in the paper that it is something that can be inferred from the breakdown moments.

Reflecting on the paper and its results at the beginning of 2026, what is perhaps more important for the conversation about the ethics of tech is what does not cause a breakdown and does not become a part of the design process or a topic of conversation between the designers. What is not there is an obvious discussion about the drones as instruments of war; the unclear purposes of our design ideation, and the established hierarchies of this design work.

The ideation work was done with some conceptual purpose of creating 'the drone' as 'the other', yet our context is totally limited to the toy drone present in the lab and the neighbouring design projects around playful drones (La Delfa et al. 2020). We did not have a discussion about the drone as technology and the network of funding around drones that allows our project to exist. We talked about the vulnerability of us, designers in a rather limited context of other playful interactions with the drone, ignoring other, certainly more vulnerable, 'users' and 'non-users' (Baumer, Adams, et al. 2013).



Figure 4.1.1: No Drone The 'No Drone' sign in Moscow, Russia. December 2025.

These taken-for-granted omissions are more important for the conversation

about the ethics of tech than what happens during the breakdowns discussed in the paper, although maybe not for the ethics of our workplace. Any context outside of what was happening in the face-to-face interaction between the designers is excluded from the workplace interactions analysed in the paper. While we certainly did care about physical safety of one another, the consequences of drones as tech were not among our matters of care. The context of drone tech was not a part of our reality at that time. This exclusion is not unique. Paper 2, which is discussed in the next chapter, and Paper 4 both show that what tech practitioners care about the matters located in close proximity to them, in their everyday context. Rarely do they engage with big tech ethics questions such as "do we need this technology in the first place" (Baumer and Silberman 2011).

4.2 Paper 2. I, We, They stances: Care and Power in Tech

Paper 2 is focused on reasoning around ethical responsibility in two main groups of tech practitioners: tech workers at the universities and in governmental agencies in Sweden. It centers on the differences in discourses on ethics and ethical responsibility between these two groups. Based on the interviews, the paper analyses how the tech practitioners in different sectors reason around the 'who should act to make sure tech is ethical' question.

These two groups of tech workers rely on very different discourses around ethical responsibility. Academic workers in our sample more often relied on a personal, individual-focused approach, loaded with emotional lexis and references to personal feelings. They see a lot of personal responsibility in their work and have little distance between themselves and the outcomes of their work. The AI professionals in government have rarely involved their personalities in the conversation about ethics. They took more distance between their work and their personality and saw themselves as part of workplace collectives, rather than as individual actors.

Theoretically, this research was inspired by the frame theory of social movement (Benford and Snow 2000). The theory focuses on the discursive work that social movements do to mobilise its supporters. According to the theory, social movements need to define a problem that has to be solved collectively. They also need to explain why this is a problem that should be addressed; define its possible solution and, most importantly for the purpose of the paper, identify the actor who should step up and solve the problem – explain why the potential supporter should be acting (Benford and Snow 2000).

4.2.1 Ethical Stances

The difference in the level of personalisation and distance is expressed in our articulation of ethical stances. Based on the interviews, we have identified three ethical stances, i.e. three discourses of attributing ethical responsibility: I-

stance, We-stance, and They-stance. The stances help practitioners answer the question 'who should act' to make sure that tech which they build is ethical: is it the tech practitioner themselves (the I-stance), the workplace collective (We-), or someone else (They-). An individualised I-stance would shortly sound as "the responsibility is mine", a collective we-stance as "the responsibility is ours", and a distanced they-stance as "the responsibility is someone else's".

An example of the I-stance is taking effort to make sure that participants of one's research project feel comfortable to take part in it, and going beyond what official guidelines would ask:

> *"There are national ethical reviews, guidelines, but it's more like the daily practice of how you're doing research. I have examples of ethical ways of working with myself, my own body. But in this case [the project discussed in the interview] it was how do you engage other people in your work and create a space where everyone feels comfortable about sharing their opinions through words, or through movements, you know, like a place where you can create trust."* (Interaction design practitioner, early- career academic, female)

The I stance is often problematic for practitioners when the need to act is identified, but the action is not taken. One of the examples we discuss in the paper is a situation where the person feels the need to voice their concerns about their company, but is not acting upon this need. This has serious consequences for the emotional well-being of tech workers, as stated in the quote below:

> *"Should I say something to anyone? And then I realised that the guy who has a company, he is a pretty powerful figure [...] and he has a lot of connections, and I didn't want to start moving things there. I mean, it wasn't my job in that case. I stayed there for three months and my whole self was destroyed, because I returned back and I started feeling that I should have said something and I didn't."* (Interaction design practitioner, early- career academic, female)

The we-stance means that the practitioner identifies with the company or organisation where they work and considers their responsibility to be collective[2]. An example here is considering that the company, as a collective actor, has a responsibility in front of another collective actor, the customer, but not distinguishing one's personal responsibility from the broader entity one is a part of:

> *"I think we also might have an educational obligation as a company. We should actually inform, in some sense, our customer*

[2] a collective does not necessarily need to be a company, but these are the examples we had in our data

> *base what it means to buy a data-driven product."* (AI practitioner, research director in a private company, male)

The reluctance to distinguish between individual and collective is even more visible in the next quote, where the interviewer was trying to problematise this possible divergence between the "I" and the "We".

> *"-In case you would experience a tension between, for example, you and your projects or your organization, what do you think you would rely on for solving that problem? For example, would you rely on your inner feeling or go to maybe some documents and guidelines and procedures or talk with other colleagues or with experts?"*
>
> *" - So, how would I go about fixing that? That's something we do pretty much every day when we run a company [...]."* (AI practitioner, research director in a company, male)

The they-stance implies that responsibility is attributed to someone else, for example, tech workers in other areas or an ethicist employed in the company.

> *"I mean our work in this project is more about the core technologies and we also support the government agencies when they build their in-house applications. But we don't take responsibility for the applications that are directed out to the citizens. That's up to the government agencies, and that's their responsibility and their ownership of those techniques."* (AI practitioner, senior researcher in governmental sector, male).

We identified several strategies that technology practitioners speaking from the they-stance used to **distance** themselves from ethical problems: (1) distinguishing between the ethical and the technical; (2) limiting the domain of ethics to interactions with end-users (as in the quote above); and (3) marking only certain areas of technology application as ethically problematic and, thus, as requiring ethical reflection. These distancing techniques also explain why participants do not feel the need to shoulder ethical responsibility. For example, tech workers will most likely agree that racial bias in AI-driven decision-making is a problem, yet they may not know who should do something about that problem, what their role is, or if they have any role at all.

4.2.2 Ethics and Power

In the paper, we consider relations between the stances and power to act within organisations. We distinguished between low- and high-scale situations, which is essentially a distinction between a situation with more power and agency, such as a university study with a small number of participants, and a high-scale situation, such as developing a commercial AI system, meant for broad adoption, with multiple parties involved.

We did not encounter cases where the implications of AI built by a company where a tech practitioner works were discussed with the I-stance. However, we have examples when the interviewee could observe the examples of privacy violation or racial exclusion happening in front of their eyes and with someone who was socially close to them, their colleagues or students, and the tech they helped to create. In these cases, when the interviewee saw their personal responsibility and an opportunity to act ethically in a face-to-face situation with visible objects of care and visible consequences, they took the I-stance.

In the paper, we speculate on the consequences of different stances for workers' well-being and the development of tech. We have focused on the emotional consequences of taking the I-stance and its consequences in the situations of lacking power, which is always the case when it comes to ordinary tech workers and high-scale decisions in their companies. When the I-stance was taken in situations where the workers did not have the power to act upon these concerns, it had negative consequences for the workers – a personal emotional crisis and/or a personal risk. Our analysis indicates that the I-stance works best in the low-scale situations, where a practitioner has the power to influence the technology they create or the outcome of their work. In the situation of high-scale technology development, accepting the I-stance can lead to a high burden on a practitioner, while not generating any positive outcome. Instead, it can lead to situations of personal crisis and, in general, worsening emotional well-being of practitioners.

We concluded that, while taking the we- and they-stances is beneficial for tech workers in a sense that it can provide sense of belonging and create a safe emotional distance to workplace, they can lead to critical questions such as 'do we need this technology at all' to never getting asked and to situations where the care for the outcomes of tech is attributed either to no one, or to those without power.

The we-stance is the most reasonable and safe for tech practitioners' well-being, but it leaves a little room for attending to the issues outside of those that would already be within organisational agendas. The they-stance attributes responsibility to people outside, or no one. It does not cause harm to practitioners, yet it can lead to the problem of detachment. In terms of driving ethical agendas, it would mean that no action is going to be taken, at least not with the practitioner.

To summarise, Paper 2 compared discourses around ethical responsibility in tech work. It aimed to connect conversations about ethics on different levels: ethics as a part of everyday work and ethics as more extended concerns over the implications of technology on its users. Our study shows that taking personal responsibility (the I-stance), which sounds positive in theory and is encouraged by value sensitive design (Friedman and Hendry 2019), is rather demanding. The I-stance, which is more often taken by tech workers who hold less power, sometimes ends up being dangerous for the tech practitioners.

Table 4.2.1: Stances, risks, and responsibility.

Stance	Responsibility	Risk
I-	high	high
We-	mid	mid
They-	no	low

4.2.3 Limitations and Future Research

A critic may point out that the distinction between the stances could be explained by the data collection structure: while the interviewees in academic environments were recruited through personal connections, the interviewees in the governmental organisations were chosen because of belonging to certain organisations and may have spoken as representatives of their respective organisations. To account for this difference we conducted a second round of interviews with government employees to pose more personally oriented questions and questions gauging about situations where participants felt a conflict between personal opinions and organisational positions. The difference in the level of personal responsibility remained.

Since the paper is based on a qualitative study, we cannot talk about frequencies based on the data. However, we have noticed in our data that people with low power in organisations and lower status in organisations (early career, female, young) tended to take a more personal approach to responsibility, as indicated by speaking from the I-stance, which would put them into a more vulnerable position and burden them with responsibility. While this remains a hypothesis to be tested in the future, our findings do suggest these hypotheses.

4.3 Paper 3. Voicing Discomfort in a Group Workshop

Paper 3 analyses verbal (voicing) and non-verbal communication of discomfort in a face-to-face workshop. The conclusions of the paper concern the possibilities and constraints of voicing problematic issues in public. We were interested in how the specific group of tech workers – designers and HCI researchers taking part in the workshop, who belonged to the same professional community – would communicate and avoid communicating discomfort in a face-to-face group interaction and in the post-workshop interviews.[3]

This paper continued to explore some of the themes brought up in Papers 1

[3]The participants were recruited through personal and public invitations. We had eight participants in the workshop, most of whom were designers and design researchers to some extent familiar with body-based design, belonging to one extended community and knowing each other, but not being close friends.

and 2. I knew from the interviews in Paper 2 that designers and researchers at universities often deal with ethically problematic issues on their own. I was interested in how these individual struggles could be shared and turned into a collective issue. I approached this workshop as an experiment in the possibilities and the limitations of sharing the uncomfortable in a group setting. In this particular workshop I did not bring up any discomfort from the interviews, although some of the problems we tried to address in other workshops were specifically related to working with pain and discomfort (Popova, Park, and Demir 2023; Chicau, Popova, and Fiebrink 2024; Popova, Chicau, et al. 2024).

The workshop was initially organised to explore the experience of **discomfort** – a concept first addressed in Paper 1. The source of discomfort that we expected the workshop participants to experience was in the workshop exercises that required touching and being physically close to other workshop participants.

We have identified the three sources of discomfort that the participants named. The first was breaching **the established rules of interpersonal interaction**, for example, touching the partner. The second was **nonverbal feedback of their partner** – the gaze, face expression, body language, and pauses. The third was what we called a **felt sense** – a feeling that something is not okay that could be perceived as discomfort, fear, resistance, or tension.

One of the main workshop goals was to address the problem of consent in body-based workshops involving physical touch and close physical interactions between people who are not trained as dancers or actors. Workshops including physical exercises inspired by yoga, breathing practices, contact improvisation and sharing experiences in a group were typical at my tech workplace (Höök 2018). I took part in many during the first two years of my PhD. I have also organised and co-led some of them to explore how interpersonal interaction can be used for stimulating conversation about ethics (Chicau, Popova, and Fiebrink 2024; Popova, Park, and Demir 2023). When I was analysing the data from the workshop interactions, I had in mind my own experience of participating in the workshops.

Some of the paper's findings concern conducting body-based workshops in an ethical way. We highlight the risks of underestimating the power imbalance between workshop organisers and workshop participants, placing too high expectations on the participants to speak up or act on their discomfort during the workshop, and underestimating the importance of social conventions around touch and physical interaction between participants.

The possibilities and limitations of voicing discomfort in a group setting have become central to this paper. We demonstrate that voicing discomfort requires interaction effort. Sharing that they did not feel comfortable with their partner during the exercise needed to be mediated by structures like following, where

critique is bracketed within positive descriptions (it was fun, but):

> *"I thought it was... thank you, it was very very fun, but I think my mind is more on negotiating the movement and what we did. And we had that, of course, initial establishment of contact and during the exercises sometimes we kinda lost it and had to reestablish, and it is not always the same next time I think. It was kinda fun."*

There was a complicated group dynamic when critique was being shared. For instance, laughter was a marker that a topic brought up by one participant was going to be supported by others. The participants also relied on mitigating narratives to soften critique, for example, saying something like 'it was fun, but'. While the workshop participants did share discomfort non-verbally during the exercises and relied on the feedback of their partner, for example, their gaze, facial expression, body language, voicing discomfort in public was different and required specific interaction work.

Although we advertised this workshop as a workshop about discomfort, the group sharing still required work. Some of the participants were more open about their discomfort in the interviews than in the workshop itself. What that means is that sharing problematic issues with colleagues informally and without established ways at hand cannot be expected by default - something that we explore more in Paper 4.

4.4 Paper 4. Making the World a Better Place? Ideology, Discipline and Agency in The Tech Sector

Paper 4 is an interview-based study exploring ethical reasoning at tech workplaces. It is based on the interviews with tech workers in European companies, although the sample is not representative of European tech workers as a population. The paper had a theoretical purpose to bridge the gap between academic discourse on ethics, especially the approach placing a lot of responsibility on an individual designer and the folk morality of tech workers that limits the role of individuals in tech ethics. Do tech workers care about ethics in their work, and do they believe that what they do is ethical? Do they think about ethics in relation to the tech they create? And how much do they shape the tech they are employed to create? The paper addresses these questions based on the interviews with 20 tech workers.

The Paper 4 relates to the work of Robert Dorschel (Dorschel 2021; Dorschel 2022a), who pointed out the contradiction between the relative autonomy of tech workers and their status as hired labour, that is, as workers rather than business owners. Since tech workers are hired labour rather than owners of the capital, they could potentially be interested in advancing a different sort of technology than the one we currently have.

#	Gender	Role	company
1	male	industrial designer	design agency
2	male	UX Researcher	design agency
3	female	software engineer	startup
4	male	software engineer	startup
5	male	tech lead	mid size company
6	male	software engineer	startup
7	male	ML engineer	startup
8	male	data analyst	startup
9	male	manager	big company
10	female	UX Researcher	mid size company
11	female	product manager	mid size company
12	male	software developer	consultancy
13	female	software developer	big company
14	female	UX Researcher	big company
15	male	software developer	consultancy
16	female	UX Designer	startup
17	female	UX Designer	startup
18	male	UX Researcher	big company
19	female	UX Designer	startup
20	female	Technical Writer	big company

Table 4.4.2: Participants by Gender, Role and Company

I also relate to the studies of Gideon Kunda (Kunda 2006), who explored the corporate structures of tech organisations in Silicon Valley in the 1990s. Kunda pointed out the contradictions between the perceived horizontality and sophisticated hierarchy of power contained in the corporate cultures of the tech companies. I found that the same contradictions as discovered by Kunda are present in the work of tech practitioners, who deal with rhetoric of 'the democratic' inner culture and the apparent inequality between different groups of workers in organisations. The "Exit, voice, loyalty" (Hirschman 2004) provides a framework for looking into the ways in which tech practitioners can act in favor of changes. The paper shows that 'exit' is the main protest option that can be taken by tech practitioners.

4.4.1 Making the Ethical Choice

The interviewees care about ethics, but consider the moment of choosing the company or the industry to work for as the key moment of making an ethical

choice. Besides accepting or declining on a job offer, the interviewees saw their ability to affect the products as very limited. They rarely talked about their work on the product in terms of ethical reasoning. It does not mean that they did not care about the product, but their perception of the limits of individual agency was quite narrow. This is how one of the interviewees described the choice:

> *" Well, it seems to me that you have agency in any case... in terms of... you can work, say, in football, but if you don't agree with what they do... why working there? if you do not agree-with what they do, why work there in the first place? What is the point then? [...] Of course, you can try to improve something from the inside, but I'm not sure. If you fundamentally disagree with a company's actions or with how it treats its employees, then I don't really see the point of working there in the first place."*(Business analyst, male, startup).

The choice of a company to work for relates to the values of a tech worker and the ideology they adhere to. Some of the tech workers would refuse to work for the military industry because of ethical reasons, while one of the interviewees was becoming more and more positive about the possibility of working for the military in Sweden. Some of the interviewees said that they would not be working in the oil industry, many would be against taking part in the mass production of clothes, as it affects the environment negatively, while for some it did not pose problems. The most conventional no-go industry was gambling. This can be because the narrative of 'doing good' and solving problems is the least applicable to gambling (Schüll 2014).

IT and engineering have a long history of maintaining the problem-solving narrative (Morozov 2014), as the profession that alleviates 'users' pain points' and, according to Google's popular slogan, makes 'the world a better place'. Although such a positive outlook has been lately criticised more and more openly, the interviews indicate that the problem-solving ideology persists. The interviewed tech workers often relied on a narrative of 'helping others' to explain the purpose of their work and to justify why what they do is ethical.

4.4.2 Doing the Right Thing

The 'solutionist' narrative of tech work is used as a default ideological framework to explain what the tech workers do. The tech workers rely on this narrative even when the 'help' may not be so obvious for an outsider. To give examples, building internal applications for HR departments in a bank or optimising data systems in the financial tech sector are not conventionally seen as 'helping people':

> *" How do I see what I do? Well, I develop products that help companies that really need money, conditionally, for financing,*

> *that can grow and become cool companies, raise money. That is, I help our investment team accept, find these companies [...] I help highlight companies that are very promising in their fields."* (Male, business analyst in a startup).

The narrative of doing the right thing is strong, but tech practitioners/interviewees also need to find justifications to counter external critiques of the tech industry (Su, Lazar, and Irani 2021). That can be a reason why some of the interviewees had to shift between the 'doing good narrative' and the 'not doing harm' or doing 'neutral' job narrative.

> *"for me it is very important, of course, yes, that what I have been doing does not influence the world in some negative way ... For example, gambling or online casinos. I can't work in these areas."* (Male, manager in a food delivery service).

The ideology of 'doing the right thing' provides a useful justification and explanation for why the situation of no change is okay. These are similar to the practices of distancing described in Paper 2. These good discourses are helpful for maintaining the status quo and explaining why the status quo is not that bad.

4.4.3 Changes from within

When it comes to introducing change from within, the channels for doing that are unclear and mostly unavailable for the tech workers I interviewed. These entry to mid-level workers largely lack the power to reinforce any changes by themselves. Additionally, the internal corporate cultures of tech companies and the established practices of disciplining largely prevent the possibilities of collective self-organisations from within. For instance, there is a culture of non-complaining that one of the participants mentioned in the interview:

> *" I've been told at one point that was being negative, because I went without a project for a while. And then like the work you do when you're without a project is kind of meaningless, but you still have to put up a face and be happy for the newcomers."* (male, software developer in a consultancy)

A culture of 'non-violent communication' can also be challenging for voicing problems, because it imposes specific rules of communicating critique and the restrictions towards the kind of critique that can be brought up. These rules have the consequence that it is not easy to share any problems between the employees.

The interviewed workers sometimes tried voicing their opinions publicly and engaged in writing collective letters, asking questions on the corporate Q&A sessions or posting openly critical messages on Slack. These issues, however,

mostly related to protecting their own benefits or their colleagues in the situations that they could relate to, such as cases on discrimination of non-technical or temp workers from permanent employees.

> "– So, from what I get, people were pretty pissed off in the comments, but nothing really changed because of it?
>
> – Yeah, exactly. I mean, you're just this powerless little person — what are you gonna do, wave a pitchfork and a torch or something?
>
> – So, you ended up quitting, right?
>
> – Yeah, yeah, I did. At some point, I just left." (female, manager in an educational startup).

To sum up, the interviewed tech workers did care about ethics, but there were a few things they could change. The interviewees, in some situations, acted in support of their colleagues and spoke up in favour of those with whom they felt solidarity, like immigrant workers dependent on a working visa. Sometimes, they did engage in collective action to protect their employee's benefits from being taken away. At the same time, speaking up did not always lead to change. Exit and choice of a different company or a different field of employment were the most powerful acts the interviewees could do.

Paper 4 was focused on the ideology of the tech work and the disciplining practices within tech organisations, which limit the practitioners' agency and their desire to act. The ideology of solutionism and the existing narratives of "helping others", "not doing harms" or "being neutral" was often expressed in the interviews, offering protection from criticism and accounts for why the tech workers could keep doing their jobs without necessarily feeling moral pressure to act[4].

4.5 Summarizing Papers

Taken together, the four papers presented a fragmented picture and examples of tech workers' reasoning around ethics in different contexts and tech workplaces. Shifting between design researchers and tech workers in the industry, the papers illustrate differences between interpersonal ethics and ethics of technology; and the discrepancy between the limits of the tech workers' individual power and the complexity of technology they create.

[4]Much like the ethical stances alleviating the emotional burden of sensing responsibility and not acting on it.

4.5.1 RQ1 What do tech workers care about in their work and their workplace?

What and who do tech practitioners care about in their work and their workplaces? I asked this question to understand whether the tech workers care about the ethics of tech in their work at all and, if so, what they mean by ethics.

We have learned from the studies 1, 2, 3, 4 that the tech workers, who participate in the study, do care about their colleagues and their users, the quality of their work, and, in general, the narratives of 'helping others', 'not doing harm', and 'being neutral' are important for them. However, they do mostly care about people and things that reside in close physical and social proximity. This is not specific to tech workers, but a more universally shared quality and the problem with morality, which risks excluding "the other" (Tronto 1993). I come back to this topic in the discussion.

The tech workers can emotionally recognise the consequences of their work in their immediate surroundings and experience them as ethically problematic. I gave one of the obvious examples of that in Paper 2, when a computer scientist suddenly noticed the problems with privacy, when noticing the relation to his colleagues. This is something that Hemmings called affective dissonance (Hemmings 2012) – when previously abstract knowledge suddenly becomes a part of one's reality through their own experience. Shilton made a very similar observation on values of privacy becoming more relevant to computer scientists, when they got to test their own devices (Shilton 2013).

> *"That particular demo machine was just outside my office, and I wanted it to be in play. And there was definitely an ethical problem to force that upon [others], because [...] There are slightly more people within computer science who are more sensitive about such issues, who are very careful about computer security and integrity and all of that."* (AI practitioner, senior academic, male)

A similar conclusion can be drawn from Paper 1. Being attentive to one kind of vulnerability of those physically close to us, we glossed over other potential harms and way more serious vulnerabilities of others. The broader consequences of technology have not entered our everyday discussions, but also other sorts of vulnerability of us as workers – social, economic and migrational differences were perhaps present in our interaction but never topicalised.

Paper 3, based on a workshop, is a good example of how the perception of ethics differs between the level of face-to-face interaction and the level of organisation. In the face-to-face workshops, the participants are essentially guided by feeling and unspoken rules. The whole interaction is based on the 'affective knowing' and tacit understanding of communicative norms. In the organisations, the affect and emotion come in (or at least mentioned) when there is a conflict, tension or a breakdown.

Tech workers whom I interviewed for papers 2 and 4 expressed that they cared for others beyond their immediate interactions, like 'helping people' or 'not doing harm', but they mostly saw their main ethical choice to be made during the moment of a job choice and the responsibility to be towards those who are socially close.

The potential of claiming technology 'neutral' or 'not doing harm' is quite broad. I had interviews with tech workers, who gradually transitioned from seeing a product they built as harmful to seeing it as 'neutral'. As I see it, the discourse of 'neutrality' (if 'there is no obvious evil, it is neutral') - can be easily adapted in the circumstances, where the possibilities of exit, voice, loyalty (Hirschman 2004), or solidarity, are not open for practitioners.

There are multiple aspects of their work, which the tech workers named as ethically sensitive in the interviews, for instance, storing data properly, making sure that users are not harmed, or collecting informed consents and not disclosing data in academia. However, they were mostly interested in discussing the issues of their own ethical decision making, the moments of high emotional intensity, breakdowns or crises, whether these were about realising the potential of their action or the need to take a difficult decision and engage in a conflict.

4.5.2 RQ2 Whom do tech workers attribute the ethical responsibility in their work?

The tech workers can attribute ethical responsibility, i.e. the agency to act ethically, to themselves as individuals, to their company as a collective entity, or to the undefined 'others' – these are the three main ethical stances that we have identified in Paper 2. These ethical stances are, essentially, **the ways in which tech practitioners reconcile the limitations of individual agency with the complexity of tech development.** The tech workers rely on these three stances as the discourses that are already there to comprehend and account for their action.

The I-stance is about taking personal responsibility. The I-stance refers mostly to the domain of face-to-face interactions, where consequences of one's actions can be experienced directly through a form of affective dissonance (Hemmings 2012) or affective attunement (Amrute 2019) experienced by a tech worker.

The we-stance is a collective stance that creates a form of solidarity between the tech workers when they see themselves as belonging to a group with a shared responsibility. The we-stance can be empowering and positive for an individual who does not have to face all the complex technical issues related to technology by themselves. In our research, the we-stance was mostly about solidarity with a bureaucratic organisation or a commercial company, not with self-organised grassroots organisations. Conformity to this type of 'we' could have a negative consequence that the questions outside the business framework or bureaucratic

framework not be asked from within the organisation, including the question 'whether we need this technology in the first place'.

The they-stance attributes responsibility outside the reach of the tech worker, to someone else, for instance, to an ethicist working in the tech company or to other people working with the tech user directly. In Paper 2, we have distinguished a number of distancing techniques, narratives within the They-stance: 'ethical is not technical', 'ethics is only relevant when the product involves the end users', 'ethics is only relevant for some domains of tech production'. These narratives protect the tech workers from the burden of individual responsibility and possible disempowerment. However, at least in our studies, the distancing techniques were often used by the people in positions of power in their specific organisations, who could have had more possibilities of driving change than those usually taking the I-stance.

This is an observation to be probed in future studies: personal responsibility for collective matters is often taken by the least powerful, and the ethical responsibility is attributed to the less powerful by the companies. Some examples from outside of my research come from the recent past with tech companies hiring ethicists, who can be fired easily, and who have a limited epistemic power in the companies (Widder 2024; Widder and Nafus 2023). At the same time, the they-stance and the position of moral invincibility were often taken by those who are most protected by company hierarchies – more senior, more 'technical', more often men than women – those who usually accumulate more power in organisations and occupy the positions of higher status.

4.5.3 RQ3 What are the possibilities and limitations for collective actors in tech workplaces?

Individual agency is quite limited when it comes to solving big technical problems. The sense of fulfilling an individual moral duty and the experience of doing so are more closely related to one's perception of oneself as an ethical person rather than to making technology that is not harmful, good for the world. What are the possibilities of forming collective subjects that would unite different tech workers to challenge the status quo?

My perspective on collective action has been inspired by the studies of social movements (Benford and Snow 2000). Specifically, by frame theory, which implies that first a common agenda must be formed, i.e., an individual ethical discomfort or 'affective dissonance' (Hemmings 2012) must be shared with others. When a shared collective agenda exists, it becomes possible to move on to a collective action. Social movements also transform individual experiences, for example, helplessness or grief, to the feeling of anger and a collective effort to change the situation (Gould 2009, pp. 200).

Papers 3 and 4 have both explored the possibility of creating a shared agenda between the tech workers. In Paper 3, we experimented with voicing problems

in artificial settings to study critique and voicing problems on a theoretical level. The reasons for discomfort originated in the workshop itself. Paper 4 was more focused on the reality of IT work and communication in tech companies and the exploration of related possibilities for unification.

Paper 3 demonstrated that voicing and sharing critique is not straightforward socially. Bringing up critiques relies on complex social orchestration, where critique needs to be presented carefully to be accepted as a shared group concern. Even within groups of like-minded people in the safe workshop settings, the development of trust and the ability to speak up did not happened automatically. Paper 4 showed that tech workers do occasionally voice problems within their companies. However, the mechanisms of delivering critique in tech companies are also sophisticated. The critique can either be delivered through the channels of corporate communication or one-on-one communication with the manager. This has been noticed by other researchers, for instance by Sarah Ahmed, tracing the institutional journey of complaints and showing that complaints often backfired on the one seeking support (Ahmed 2021).

Sometimes problematic issues are shared between co-workers and voiced via public channels of communication, e.g. on Slack or in a collective letter. However, these issues in my studies revolved not around the social consequences of technology, but the workplace problems that bother the tech workers, such as reduction of their benefits or inequality within the company (Paper 4, Amrute 2019).

Even in situations when the problems are voiced, there is not much chance of implementing change. In my data, I have some examples when tech workers did speak up and act to help others, but they were relatively small (negotiating exit conditions for immigrant workers) and related to internal issues (preserving benefits for tech workers). In more high-stakes situations, such as layoffs, the tech workers did not have the collective power to keep the co-workers' jobs. People going through layoffs or conflicts with their companies had to go through conflicts by themselves, with only the moral support of their colleagues, at best. There are exceptions, but these were the observations in my data set.

5 Discussion

This thesis is an exploration of ethical reasoning in the context of technology development. There are four papers included that originate from two research projects. Papers 1 and 3 are positioned within first person design methods (Lucero et al. 2019), while Papers 2 and 4 have a more sociological orientation, being based on the interviews with the tech workers working in academia, industry, and the governmental sector. As a consequence, my work relies on very different methods to explore the ethics of tech workers and technology workplaces. Working on the intersection of research traditions, I suggest a perspective on ethics grounded in the experience of the tech workers, which will hopefully resonate with some of the researchers studying technology within the field of CSCW. The purpose of this chapter is to connect my findings with the literature in relation to the three topics important for this thesis: emotion and feeling; tech workers' collective power; and, finally, the meaning of ethics in relation to tech development.

5.1 Emotion, affect and ethical reasoning

> *"To be emotional means to have your judgements affected: it is to be reactive rather than active, dependent rather than autonomous."* (Ahmed 2014, pp. 3)

One of my key contributions is highlighting the importance of emotions, felt, and the 'felt sense' (Varela 1999) in the conversation about the ethics of tech work. Emotions have become a part of the discourse on ethics in HCI (Amrute 2019; Garrett et al. 2023; Rattay et al. 2024; Su, Lazar, and Irani 2021), but they still remain on the fringes of our understanding of ethics. I contribute to the studies of emotion in relation to ethics by arguing that, *when it comes to*

morality and ethics, emotions are important not as much as individual affects, but as collective experiences that are articulated in individual terms.

The papers in this kappa discuss morality and ethics through the experience of tech workers. Paper 1 focuses on *breakdowns* and the associated experience of *discomfort*, where discomfort works as an umbrella term pointing towards the experience of disturbance and unease (Chicau, Popova, and Fiebrink 2024; Popova, Park, and Demir 2023). Paper 2 focuses on ethical responsibility and the emotional consequences of different ethical senses, especially the high emotional burden of the individual responsibility of the I-stance. Paper 3 digs deeper into the topic of discomfort, the possibility of voicing individual discomforts and making them a part of the collective agenda. The paper also explores how the felt sense and experience are central for coordinating with others during face-to-face interaction in a workshop. Paper 3 explores ethics in a very specific setting, but I try to rely on these observations in order to generalise on the mechanisms of sharing the 'uncomfortable' (voicing critique) in a group setting in general. Paper 4 connects the emotions and experiences of tech workers at workplaces with the opportunities for collective action and talks about the ethical choices made by tech workers.

While there is quite a lot of work in HCI that either centres on experience (Balaam et al. 2019; Höök 2018; Lucero et al. 2019; Wright and McCarthy 2010) or emotion/affect in relation to ethics (Su, Lazar, and Irani 2021; Rattay et al. 2024), the felt affective dimensions of ethics are still far from being central for the discussion on ethics. In one of the reviews I received for the yet unpublished Paper 4, a reviewer pointed out that the focus on emotion 'fundamentally misunderstands ethics as a matter of affect (individual affect)'. I bring this up as an illustration that a more established discourse on ethics remains grounded in the separation between emotion and reason, where affect is seen as a potential contaminant to be excluded from the considerations of morality.

This discussion is not new: the danger of 'sentiment' has been a part of the discussion within the moral philosophy since the 18th century (Tronto 1993). When Tronto presented her theory of care ethics with the emphasis on context, responsibility and interdependence instead of universal justice, autonomy and rules, she also presented the contestation between 'the moral sentiment' and the supposed universality of rational judgement as a central thread in the philosophy of Scottish Enlightenment. Reconstructing the history of moral philosophy, she outlines how the philosophers of the Industrial Revolution and the emerging capitalist epoch worked towards removing the 'sentiment' (emotion) from their consideration of moral reasoning. Sentiment ceized to be a solid ground for morality with the growth of social distance between individuals in early capitalism (Tronto 1993). Moral sentiment, emotional connection, and empathy were not fit to be part of ethical reasoning, because *those who were more privileged did not feel sympathy for those lower in status.* At the same time, moral philosophy did not manage to discard the sentiment entirely. The

reliance on sentiment became the feature of private life, which was more associated with females as 'naturally' caring creatures (Tronto 1993).

The displacement of sentiment and emotion as qualities of femininity and private life instead of politics is problematic for many reasons (Tronto 1993; Ahmed 2014; Federici 2004; Hemmings 2005). An alternative, which Tronto suggests, is a moral theory combining the principles of justice and human ability to recognise and attend to the needs of others – those abilities that show up in caring practices, central to human development, like raising a child or taking care of the sick. The problem Tronto and others, including Hannah Arendt, are trying to solve is about *extending the boundaries of morality to include 'the other' – the less privileged, the one who is socially distanced from the center of power and remote from the considerations of morality*. The groups excluded from considerations of morality have varied historically: from slaves in ancient Greece, to women, immigrants, and LGBTQ+ communities. A common response to the problem of moral responsibility—when one lacks the power to undo others' wrongdoing—is to strip others of their humanity and, consequently, the possibility of empathizing with them.

> "[T]he idea of humanity, purged of all sentimentality, has the very serious consequence that in one form or another men must assume responsibility for all crimes committed by men, and that eventually all nations will be forced to answer for the evil committed by all others. Tribalism and racism are the very realistic, if very destructive, ways of escaping this predicament of common responsibility" (Tronto 1993, 59, citing Hanna Arendt).

I find this answer similar to how the problem with responsibility and morality is approached in technology development. The complexity of outcomes and social stratification of consequences is enormous in the case of tech development. Those who develop technology and those who face its problematic consequences are separated socially and geographically; the consequences are extended in time and distributed unevenly. Moral conflicts are *felt* when technology is affecting those who are socially close, but not when the effects of tech go beyond that. As identified in Paper 1, where drone technology was not problematic for our work at all, the remote consequences of technology are not experienced at the site of idea development. Paper 2 has also demonstrated that government officials working with AI to solve 'purely technical' tasks put rather strict limits on the extent of their responsibility, and so do tech workers in the industry, whom I interviewed for Paper 4. AI practitioners and tech workers may *feel for* and *care about* the experiences of their colleagues, but do not accept the responsibility for more distant consequences of technology in Papers 2 and 4.

Papers 2 and 4 indicated that the moments of affective dissonance (Hemmings 2012), that is, experiencing empathy and remorse by those occupying more privileged positions, were related to observing the effect of technology on their

peers or those who the tech workers could relate to (for instance, noticing how other computer scientists are concerned by face-recognition technology). Yet they have not experienced the tech consequences that are extended, such as deskilling or creating environmental damage.[1]

I read Tronto's version of the history of moral philosophy with the conclusion that it is not possible to entirely remove sentiment, personal and contextual, from moral theory, but relying only on the sentiments of those who are privileged and occupy power positions is not beneficial for the development of a moral theory. Hence, paying attention to the sentiment is valuable when this is the sentiment of those who occupy lower positions of the socioeconomic hierarchy. This is especially relevant when it comes to negative 'discomforting' emotions related to living through oppression. The caveat here is that it is exactly the experiences of those who are most affected and those who are most underprivileged that usually get dismissed (Toole 2019).

Sarah Ahmed writes in the introduction to The Cultural Politics of Emotion that, in the Western context, 'To be emotional means to have your judgements affected: it is to be reactive rather than active, dependent rather than autonomous' (Ahmed 2014, pp. 3). Toole writes that those occupying the most underprivileged positions often both have the greatest knowledge of the effects of structural oppression and the least amount of epistemological resources, access to legitimate language, to talk about this oppression (Toole 2019). My argument would be that this kind of reasoning is also accepted in the fields of technology development, where people experiencing the worst consequences of tech – drone strikes, environmental disasters, and deskilling – have the least representation in the discussions on tech ethics.

The distribution of affect and knowledge within tech companies reflects the inequality outside of tech. The voices of those who occupy lower-status positions are also heard less. As an intermediate result, I can formulate a conclusion that tech workers occupying a lower status in organisational hierarchies experience the 'discomfort' in the moments of breakdowns, when they face the dilemma of feeling the need to act and not being able to do so. Those occupying more senior positions may rely on various discourses of moral justification for 'distancing' themselves from questions of morality, for instance, that *ethical is not technical* and the *technology is neutral*. This idea may work as a lens for rethinking the unequal distribution of moral affects among the tech workers.

I started my exploration of ethics within the first-person perspective (Balaam et al. 2019; Höök 2018; Lucero et al. 2019) as part of the experience-centred design in HCI (Wright and McCarthy 2010). Paper 1 and Paper 3 were written with this theoretical approach in mind and mostly aimed at contributing to this specific domain within HCI. Reflecting on these studies now, I think that

[1]Facing here does not necessarily mean seeing, but being able to incorporate the experience. Distancing techniques can be useful to deflect the experience even when it is faced.

focusing on emotions in relation to ethics is important because emotions move between the boundaries of the individual and the social (Ahmed 2014). Because of that, when it comes to ethics and morality, emotions are interesting in light of their *relation to power* , which still remains a fringe object of study for HCI with some important exceptions (Dombrowski, Alvarado Garcia, and Despard 2017; Fox et al. 2016; Irani et al. 2010).

Based on the papers collected in this kappa, I emphasise that moral affects are not individual. The emotions may be experienced individually, but the reasons for them are often structural, such as in the situations of a moral conflict between 'doing the right thing' and preserving one's own well-being, described in Papers 2 and 4. The value of affect as a theoretical construct resides in connecting individual, private experience, with structural problems of power distribution (Ahmed 2004; Ahmed 2014; Hemmings 2005).

The quality of emotions to move between different epistemological levels is what Sarah Ahmed discusses as the 'political economy of emotions' (Ahmed 2014). She is focused on the emotions as crossing different epistemological levels, especially the darker feelings of hatred and disgust. Discrimination does not only exist in statistical metrics, but it also resides in everyday life, for instance, in superficially innocent questions ('where are you actually from?'), and the experiences of those who encounter racism in their everyday life (Ahmed 2014, pp. 59). It is felt individually, but has structural oppression at the core.

Focus on emotion allows us to go beyond the usual distinctions: we can see powerlessness and discomfort experienced by individuals occupying similar structural positions. For instance, entry-level workers in organisations who are faced with their own powerlessness in situations of value conflict with an institution may have similar distressing emotions. Focusing on individual emotional experiences, however, can also work towards reinforcing the status quo and. Talking about feelings is partly legitimised in organisations, but it is also very depoliticised and individualized. Following feminist studies of affect (Hemmings 2005; Toole 2019), I would argue that emotions emerging in moral breakdowns are the experiences shared by people who occupy structurally similar positions – such as the experiences of powerlessness by those taking the I-stance in Paper 2.

The papers in this kappa acknowledged the importance of emotion in relation to ethics, but also their limitations. There are quite a lot of limitations and problems in the discussion on responsibility in tech development coming from the ideology of technosolutionism, underlying the positive role of the engineers as the ultimate creators of better worlds. These are the discursive strategies helping tech workers to distance themselves from the implications of technology in the world (the examples of these discussed in the paper would be "ethical is not technical", "ethics only becomes relevant when we build tech for the end users", "ethics is only relevant for specific fields"). The distant consequences of technology are not felt; they become foregrounded only when they are experienced

in relation to socially close ones – like a colleague or a student, as we discussed in Paper 2.

In conclusion, I want to account for the need to talk about emotion as part of ethical reasoning. When ethics is separated from individual reasoning and sense of morality, it turns into laws and regulations – the domains of professional expertise of lawyers and bureaucrats, a matter of compliance (Orr and Davis 2020), rather than a space for making an ethical choice.

The fact that people in lower power hierarchies experience moral breakdowns and talk about the situations of power imbalance with the vocabulary of individual emotions can also obscure that what is at stake is a power conflict. Individual emotions could create a ground for collective change – this is what Deborah Gould shows is done by social movements. She studied the movement against AIDS, showing how it helped to change individual grief into collective anger. This is something that could be imagined in organisations of tech workers, yet the way tech is organised makes it rather unlikely – something I address in the following subsection (Gould 2009).

To summarise, when it comes to morality, individual experiences have structural reasons. Care ethics (Tronto 1993) and feminist epistemology (Toole 2019) suggest directing attention to the experiences of those who occupy relatively underprivileged positions in tech organisations (obviously, besides those who experience the consequences of technology) rather than those who have more access to power. Power and knowledge can and often do overlap (Foucault 1980); those who are disempowered have more access to knowledge on oppression and inequality, yet often lack epistemic tools such as language to talk about oppression (Toole 2019). Having space to attend to the experience of those who lack power is needed for developing ethics that combine ethical principles with individual freedom and agency to take responsibility for creating technology worth building.

5.2 The tech workers

In this subsection, I look at the possibility of collective action and political mobilisation among the tech workers, building on the interviews collected for papers 2 and 4 and the studies done by other scholars. The question that I want to explore in this section is whether tech workers can act as drivers of change and resistance from within their workplaces, and whether tech workers could push towards the development of different kinds of technology.

Dorschel, who studied the subjectivity of tech workers, suggested that tech workers do have potential for collectively influencing the shape of technology (Dorschel 2022a). Given that tech workers have economic power, relative independence and autonomy, and direct influence on the shape of technology being produced, they occupy an intermediate place in the class hierarchy of high-tech capitalism (Dorschel 2021). Dorschel suggested that tech workers may have collective

power to affect the shape of the tech they are building:

> "Like Irani (2019), I find that tech workers embody an honest longing for political purpose and world improvement that runs deeper than notions of 'solutionism' invoke. Tech workers talk a lot about the communal responsibility of their craft with regard to their inscription power. " (Dorschel 2022a)

In papers 2 and 4, I looked into the possibilities of individual and collective action within the tech sector. In short, these papers suggest that tech workers have limited power and opportunities for resistance at their workplaces. Most of the collective efforts of tech workers in my interviews were directed at changing their workplace conditions rather than technology – this is similar to what, for instance, Amrute discovered in relation to Microsoft workers (Amrute 2019). Even these efforts to preserve workplace benefits, such as the right to work remotely and maintain autonomy from the bosses, were often not successful for the tech workers I interviewed. The same is true for efforts to push toward a more just distribution of benefits between engineers and non-engineers in tech companies.

One of the reasons why it may be difficult for tech workers to get engaged in a collective action is that they are quite diverse as a population; they mostly work individually[2], and they do not tend to stay in one company for too long (Dorschel 2021; Amrute 2019). Tech workers have different cultural backgrounds; they are often dependent on migrational regimes, which makes them more dependent on the employer (Dorschel 2022b).

There is also a lot of inequality between different professions within tech, for example, between data scientists and UX writers (Paper 4), or between designers and engineers. Engineers working with hardcore technology typically have more weight within tech organisations, at least when it comes to economic power and autonomy, than those who work with users, like UX researchers or marketing specialists. This observation resonates with Gideon Kunda's earlier research on engineering cultures in the United States (Kunda 2006).

Kunda described how, despite the proclaimed horizontality of tech companies, there is a lot of inequality between different types of tech workers (Kunda 2006). Within a specific engineering system of values, less technical and more people-oriented professions, such as UX writers and UX researchers, are seen as less sophisticated and less important for the core business. They are less protected from layoffs and sometimes have less access to company benefits than those who possess the 'hardcore' technical skills, especially AI engineers and software developers (Paper 4). This is the distinction that is recognised from within the companies, both by those who have more and less status, as follows from the interviews for Paper 4.

[2] Individual or collective work depends on a role, for instance, software developers may have more collegiality than the UX researchers.

In terms of the discussion on ethics and responsibility, it means that the tech workers, who interact with users and take up roles of user advocates, have less power than engineers, who work with 'core' technology and 'objective' metrics, like those developing ML models. The same is true for hired ethicists. Here is how David Widder describes engineering elitism in relation to the AI ethicists, who were hired to ensure ethics in the big tech companies:

> *" the work of AI ethics is often understood as a "subjective" endeavor, and thus within the technology cultures in which this work sits, is cast as lower status when compared with the engineering work, seen as objective or rational"* (Widder 2024).

When rationality, objectivity, and remoteness from the user are given high status, it can be difficult to identify ethical issues based on individual concerns or acknowledge the political nature of technology, and the possibility of discussion and disagreement about technology worth building (Seaver 2021). Since speaking up 'subjectively' goes against engineering culture, references to guidelines can help to advance some ethical positions within tech. The problem with that, which Madaio and Wong have identified, is that guidelines are still often not used in the companies (Madaio et al. 2020; Wang et al. 2023).

A focus on objectivity also does not sit well with taking personal responsibility for the development of technology. Voicing concerns is not straightforward in general, even when the stakes are low and power differences are insignificant (Paper 3). Voicing problems is even less straightforward in companies, where people do not work together on a regular basis, do not know each other that well and do not trust one another (Paper 4).

I assume that voicing individual concerns is even more problematic when more weight is given to numerical metrics than subjective opinions. Given that the employees who interact with the users more often also have less weight in the company, they are facing more problems when voicing concerns and problems. Besides the obvious benefits, ethical training for designers and UX professionals also has risks of being harmful for individual practitioners. They may find themselves to be powerless and isolated from other workers if the amount of responsibility is not balanced with power within their respective organisations. The 'doing the right thing', 'not harming', and 'helping others' narratives serve as powerful explanatory tools, preventing individual workers from moral breakdowns and the sense of discomfort from observing the negative consequences of tech. What happens when/if tech practitioners identify ethical problems with technology? There could be possibilities of voicing concerns, exiting, loyalty (Hirschman 2004) or organising and acting collectively. Based on papers 2 and 4, I discovered that exit remains the most likely option for the people I interviewed.

Tech workers mostly notice the problems in close proximity to them (Shilton 2013). The most obvious example of these problems in my data were the wors-

ening work conditions. As hired employees, tech workers are going through the same cycles of deskilling and losing autonomy and privilege, as other workers have been through historically (Braverman 1974). Recent layoffs in the tech sector is one of the markers of this dynamics. Some of the interviewees in my studies have been through layoffs and cuts of the benefits in their companies, but there were few successful examples of resistance from within. It is possible that the diversity of working conditions within the profession limits the possibilities of creating a shared collective agenda between different groups of tech workers.

To summarise, the tech workers whom I interviewed have a strong ideology of technosolutionism (Morozov 2014), neutrality and inevitability of technology, and a number of narratives to enable distancing from responsibility for the outcomes of technology (Paper 2, Orr and Davis (2020)), as well as the narratives glossinh over the political nature of technology, promising to reconcile values and avoid having any kind of political discussion (Seaver 2021). That does not mean that tech workers are not striving for the needs of their users and or improving tech and that there is no resistance within tech (Widder and Nafus 2023; Powell et al. 2022). Yet the collectives, in which tech elites are organised and the hierarchical principles within work towards maintaining the status quo. The fragmentation and status differences among tech workers (Kunda 2006) and the elitism of the profession (Burrell and Fourcade 2021; Farrell and Fourcade 2023) together with disdain for 'subjectivity' (Widder and Nafus 2023), shape the community and hinder options for voicing problems and engaging in resistance.

5.3 What do we talk about when we talk about ethics?

"The problems with AI are not technology problems. They're problems that arise because of the political and economic incentives shaping the development of the technology."

Shannon Vallor in the interview to Sherbini (2023).

What does it mean to talk about ethics in relation to technology development? In HCI, ethics is often discussed in terms of following guidelines or dealing with unintended consequences of technology development. An example of an unintended use is an app built to protect kids from cyberbullying that was actually used for committing cybercrimes (G. Pillai et al. 2021). Another example is a platform designed to connect friends that has started to be used for cyberstalking.

Ethics can be discussed in terms of following ethical procedures, such as going through the university's ethical approval process or storing data in the correct manner. Ethics can also be discussed in terms of making an individual contribution in perpetuating inequality (Crawford 2021) and reinforcing structural oppression (Eubanks 2019).

Viewed from the perspective of structural inequality, racial bias built into systems of automated decision-making in the US (Eubanks 2019) or environmental consequences of large language models (Bender et al. 2021) appear to be a logical outcome of tech development, rather than an unfortunate accident. Viewed from this perspective, the problems with LLMs are not so much about individuals receiving wrong information or cheating during exams, but of harvesting users' data, perpetuating surveillance and reinforcing certain ways of existing in the world (Sherbini 2023) that may not be beneficial for most people. In my study, I tried to understand how tech workers reason around their responsibility for these broader implications of the tech they are building, and how the everyday work practices overlap with the consequences of tech.

There is quite a bit of critique around the use of the word ethics in relation to technology development. Some of the authors criticise the very fact that the discussion spins around ethics instead of power – why talk about ethics, if we actually need to have a conversation about power? (Widder and Nafus 2023). Why talk about ethics in relation to tech development, if the word 'ethics' refers to self-regulation, the moral law within, when technology is produced through the accumulation of power and resources? Take the example of LLMs, they consist of algorithms built by generations of mathematicians, rely on the accumulation of data and physical infrastructure for its maintenance, hardware and materials – in short, the accumulation of people, resources and the coordination between them. What does the accumulation of resources have to do with ethics, a practice of self-regulation?

One of the issues I approach in my PhD is the distinction between the ethics of interpersonal interaction and everyday work and the structural consequences of tech development. These two did not seem to point at the same direction. Ethics as self-regulation is understandable within interpersonal interactions with visible outcomes of one's actions, but it becomes something significantly less tangible in the domain of remote consequences of tech (Paper 2, 4). Ethics of interpersonal interaction can be extended to interaction with users – and this is a part of ethical training that is already established in HCI education around dark patterns (Colin M Gray et al. 2018) and bias (Hanna et al. 2025). However, the more extended structural consequences of tech that are evoked by political and economic incentives (Sherbini 2023) are difficult to grasp through the ethics of interpersonal interaction. The structural consequences of technology may be better understood by thinking about the distribution of profits and externalities. It brings us back to the central question of ethics: who gets to be included in the category of human (Lloyd 2015) – the ones who are not supposed to be harmed by technology.

The 'solution' I see for reconciling the ethics of self-regulation with the ethics of structural consequences is to attend to the power hierarchies within the tech sector itself. Technology, as we know, is not neutral; it reflects the values of its creators (Winner 1980; Latour 1992). My argument is that technology also

reflects the systems of oppression and inequality underlying the process of its creation. The distribution of power within the tech sector can then be an attainable focus of ethics from within the tech sector. Addressing the power relations and structures of oppression within the tech – something that researchers, who study, for instance, gender differences in engineering education (Borsotti 2018) already do, but that is not always defined as relevant for ethics of technology – can be a way to both localise ethics, making it observable and actionable *within* the tech sector and attend to the structural consequences of tech development.

6 Conclusion

This thesis is about ethics in tech development. It explores ethics in the work of technology practitioners, combining methods from design research and sociology. The four research papers and the kappa examine the tech workers' ethical reasoning as a combination of practices, discourses and affects. In these papers, I approach ethical reasoning as a phenomenon that is observable and empirically accessible through a combination of interviews, observations, and workshops. Theoretically, my approach to ethics is based on the ethics of care (Tronto 1993), ethnomethodology (Livingston 2008) and moral anthropology (Zigon 2007). The four papers combined in the thesis looked at different types of tech work in Europe, from HCI research at the university to the tech workers in the commercial companies.

I focused on the three research questions: what do tech workers care about, whom do they attribute ethical responsibility to, and what are the possibilities for collective action in the tech sector. Shortly, the answers are as follows: the tech workers care about their immediate surroundings, including the people they can relate to, those who are located close to them socially and geographically. Ethical issues do become more noticeable and observable when the tech workers can relate to people who are affected by tech (Shilton 2013). This, however, is problematic, since tech workers belong to the elite (Burrell and Fourcade 2021) and are usually quite remote from those who experience the worst consequences of technology.

The tech workers rely on a variety of discursive strategies to distance themselves from responsibility (Popova, Figueras, et al. 2024). Partially, this is done to alleviate the burden of individual responsibility, which comes from facing the discomfort of recognising ethical issues and not being able to act on it. Interestingly enough, distancing strategies are often used by those who occupy

higher status roles in tech companies, while employees with less power often tend to take more personal responsibility for ethical action (Popova, Figueras, et al. 2024). Overall, the kappa and the papers take a critical stance on individualising ethical responsibility in tech. I argue that this individualisation can be disadvantageous for those occupying lower-status positions in tech companies, as they often bear the burden of responsibility without the power to act on it (Widder 2024).

The problems with technology are political in nature, so they must be addressed collectively. What hinders the possibility of collective ethical action within the tech sector is that it is rather fragmented and elitist (Burrell and Fourcade 2021), and there is quite a bit of status hierarchy within the sector nowadays (Widder and Nafus 2023) and historically (Kunda 2006). In that regard, the thesis argues that, if we wanna approach the ethics of tech as an observable and actionable topic, we may address it through status hierarchy within the sector itself – something that researchers and activists working against gender discrimination and colonialism in tech are already doing (Borsotti 2018). The assumption here is that, as technology embeds the values of its creators, it also embeds the conditions of its creation, including power hierarchies and systems of oppression and exploitation. The oppression is certainly not limited to the problems of tech workers, given the long chains of production involved in tech development (Crawford 2021), but looking at oppression within the tech sector can be a step towards addressing structural problems that are local and present in everyday interactions.

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